

Time Series Forecasting

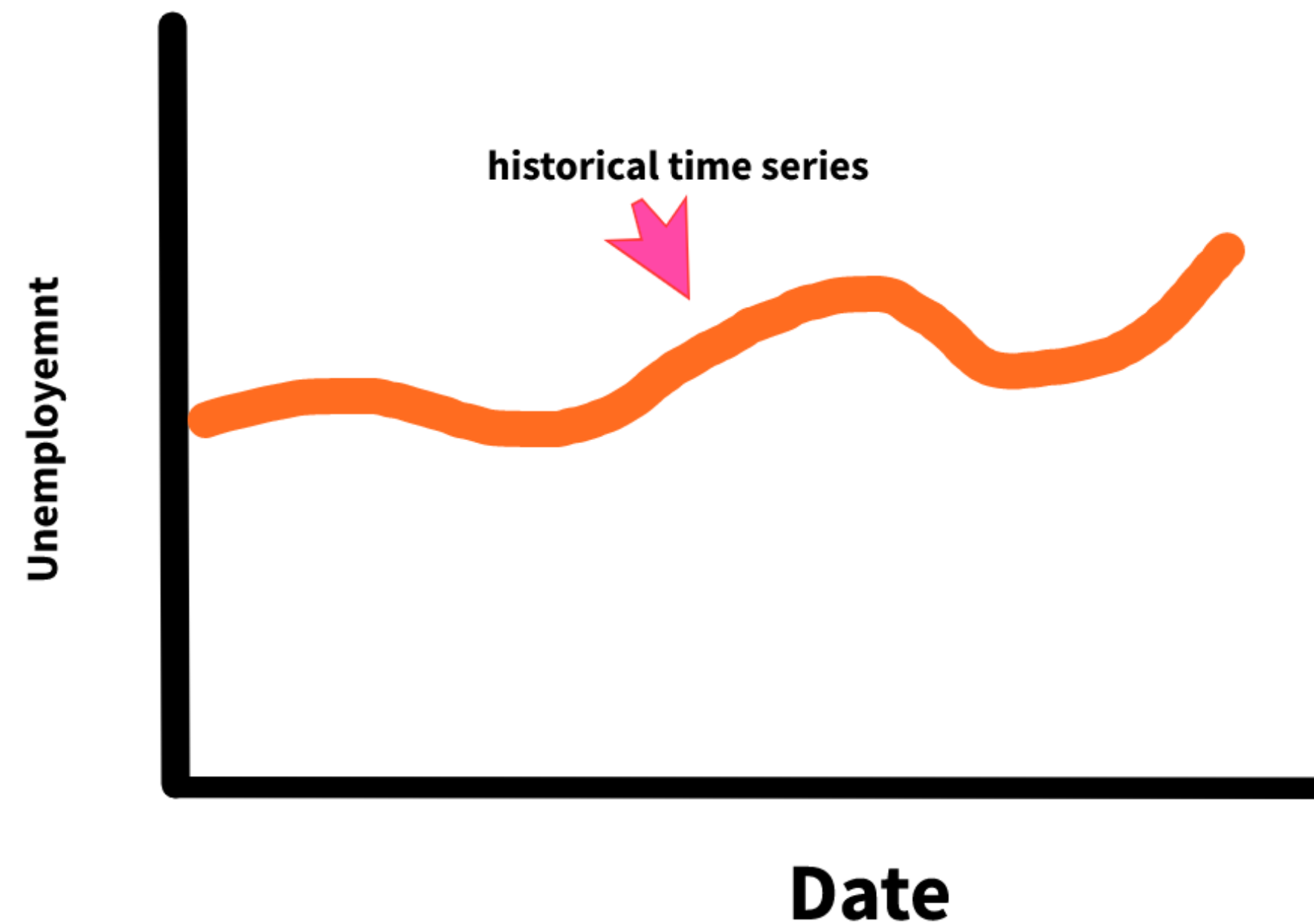
A Gentle Introduction

Brandon Mossop

The Basics of Time Series Forecasting

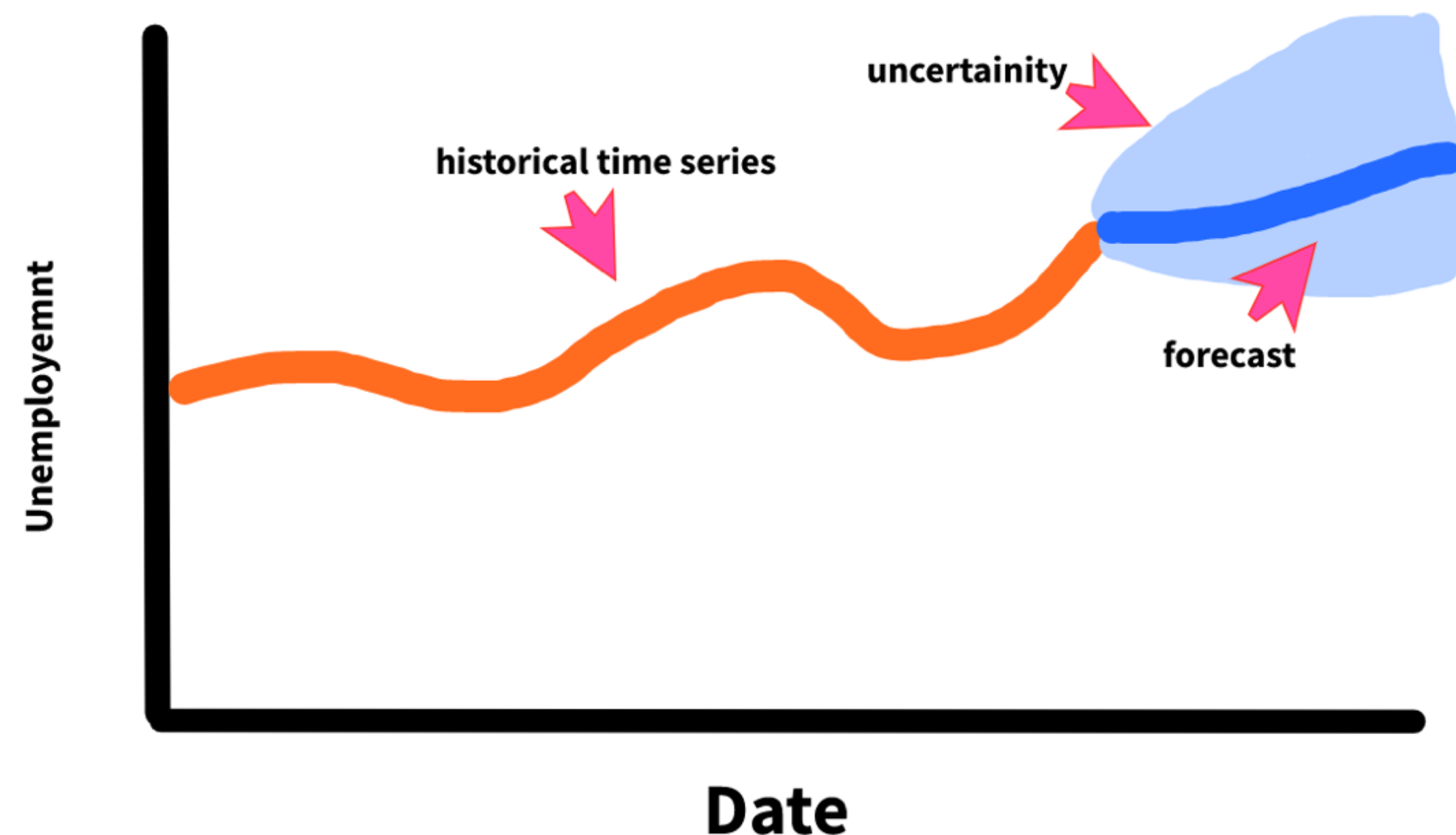
What is a Time Series?

- A *time series* is any set of observations ordered by the passing of time.
- Time series can be univariate or multivariate
- The *frequency* of a time series is the number of observations if you consider the natural time interval of measurement. For example, if you measure some variable once a day, then you have a daily frequency time series. Other common frequencies are hourly, weekly, monthly, annually.



What is a Forecast?

- A forecast is any statement about the future.
- We can forecast a single variable or many variables at once
- Every forecast has a certain forecasting horizon, the number of time steps being predicted.
- A forecast is never entirely certain, so it is essential to quantify this uncertainty when forecasting.



How Do We Forecast?

- We forecast by extrapolating patterns from the past
- The success of any forecasting method requires that:
 - (a) there are regularities to be captured;
 - (b) those regularities are informative about the future;
 - (c) the proposed method captures such regularities; yet
 - (d) excludes distorting non-regularities (noise)

$$y = \textit{signal} + \textit{noise}$$

$\hat{y}$

$$\hat{y} = \textit{signal}$$

Why should I care about Forecasting?

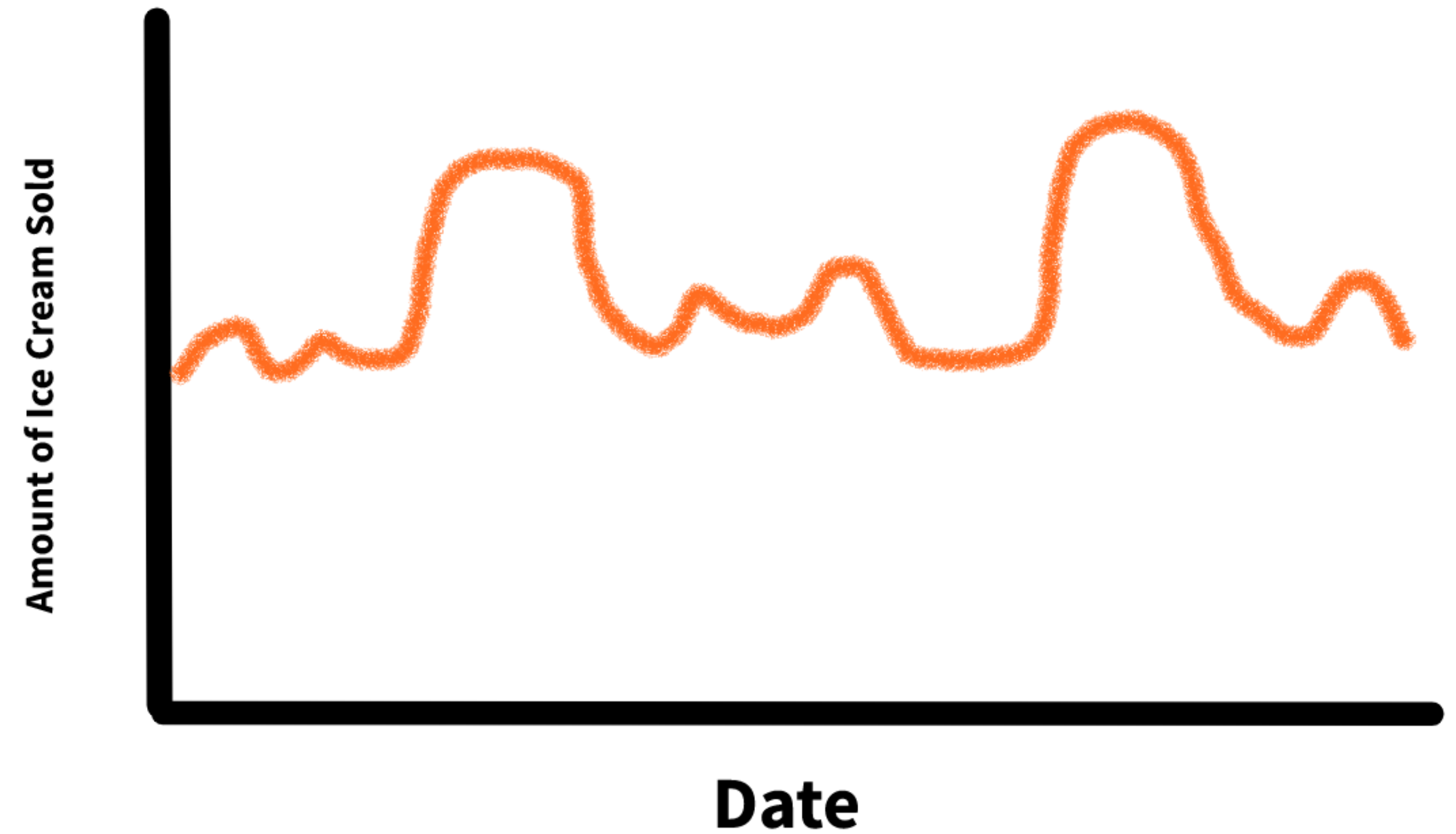
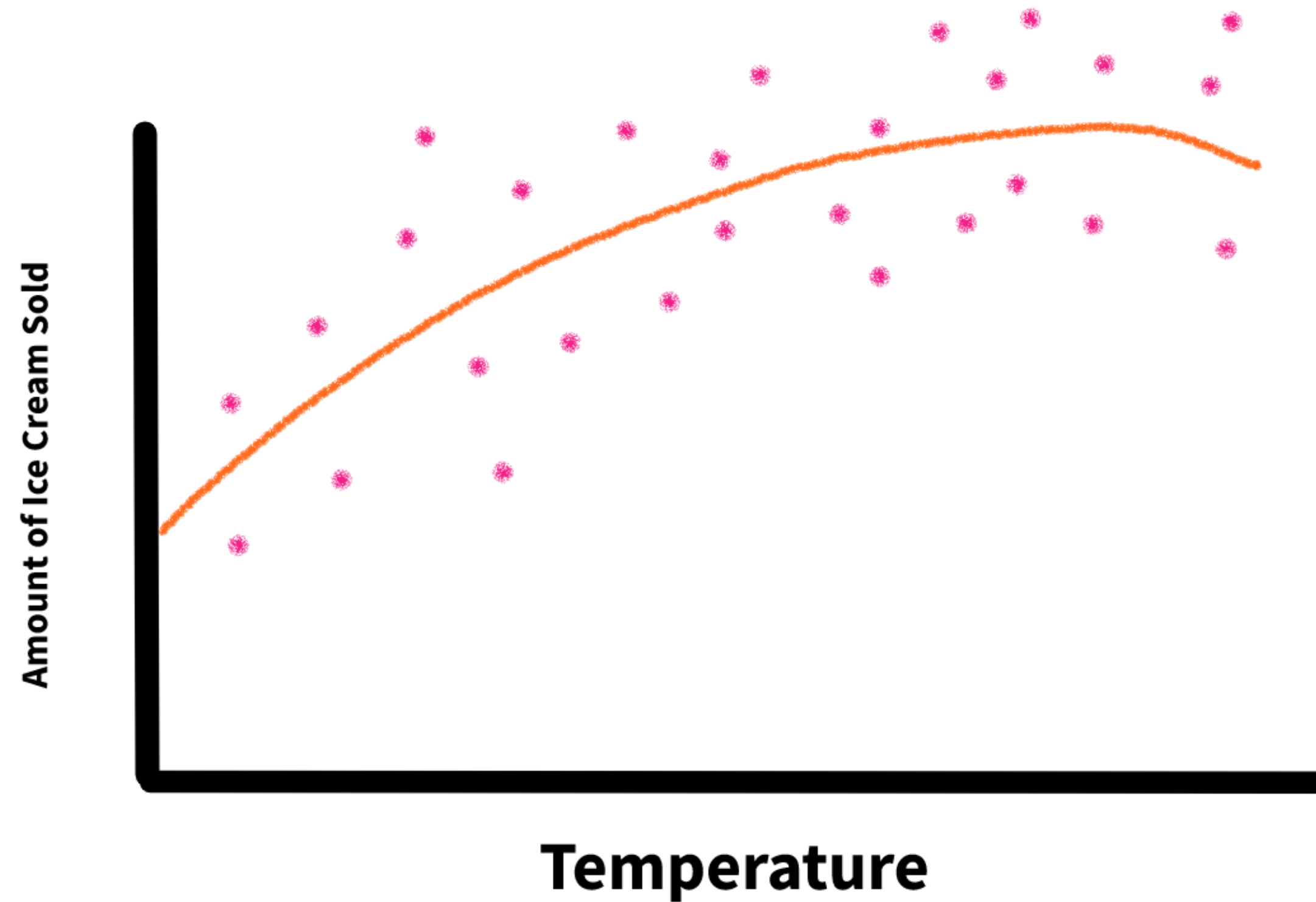
(Or the importance of forecasting)

- We need forecasting whenever we want to predict the future!
- Forecasts are needed for all walks of life for facilitating decisions: how long will this car trip take?; should I buy a book on this website or look further for a lower price?; will it rain tomorrow?
- Many of the most controversial and important questions arise from time series forecasting (e.g. what will be the rate of polar ice disappearing each year?)
- Question: Consider an Uber driver who expresses skepticism about forecasting being relevant to his work. Can you think of why he is wrong?

Examples of Time Series Forecasting Problems

- What are the expected sales for a brand of ice cream at a grocery store?
- What is the resale value of a vehicle three years after purchasing it?
- What is the future electricity load in an energy supply chain?
- What is the spread of COVID-19 infections?
- Can anyone name some other time series forecasting problems?

Difference between Interpolation and Extrapolation



Transforming Time Series Data to a Supervised Learning Problem

Date	Total
1986-01-01	9034
1986-02-01	9596
1986-03-01	10558
1986-04-01	9002
1986-05-01	9239
1986-06-01	8951
1986-07-01	9668
1986-08-01	10188
1986-09-01	9896
1986-10-01	10649
1986-11-01	8917
1986-12-01	8196
1987-01-01	10768
1987-02-01	12220
1987-03-01	14463
1987-04-01	12944
1987-05-01	11001
1987-06-01	11000



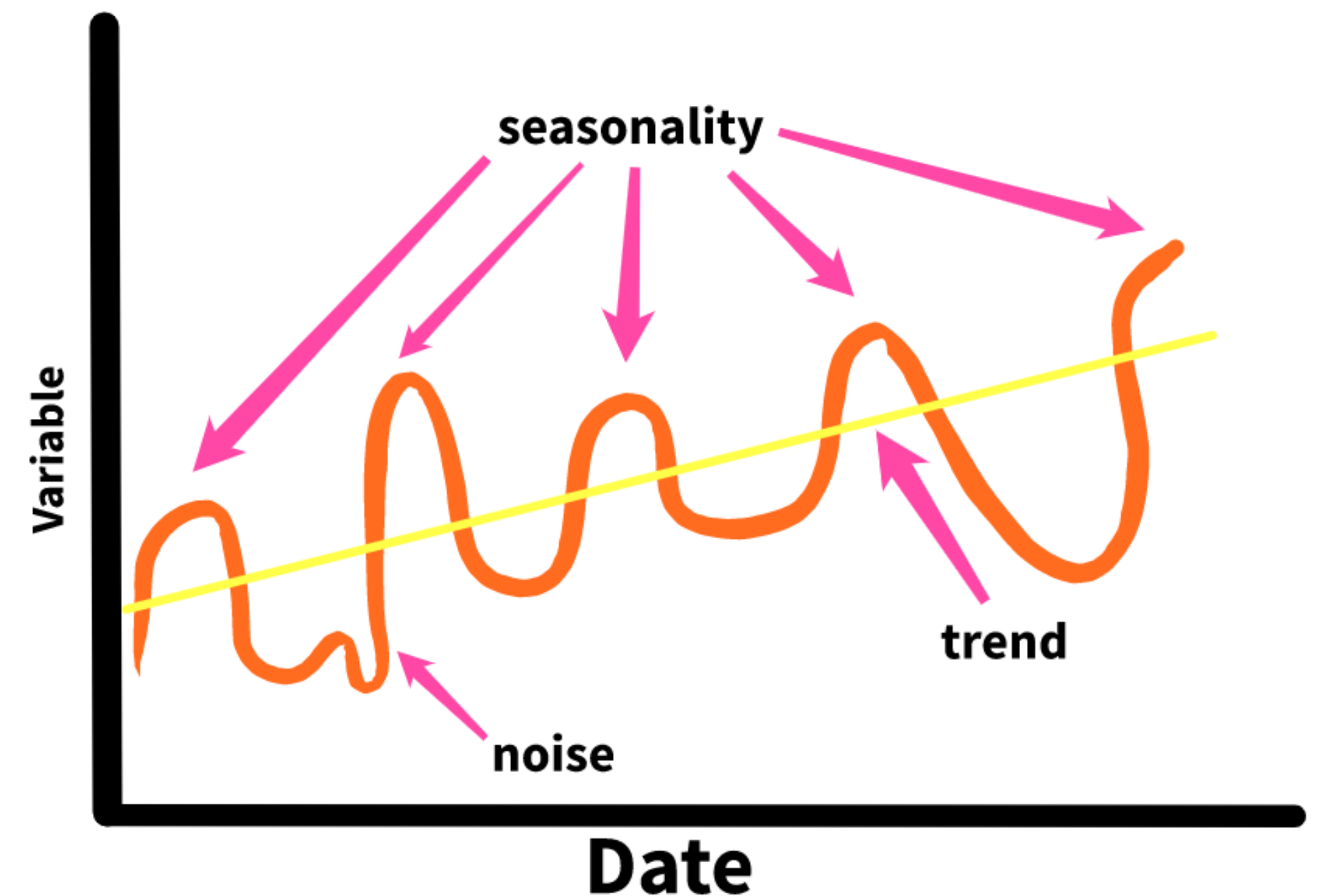
Inputs				Outputs	
t-3	t-2	t-1	t	t+1	t+2
9034	9596	10558	9002	9239	8951
9596	10558	9002	9239	8951	9668
10558	9002	9239	8951	9668	10188
9002	9239	8951	9668	10188	9896
9239	8951	9668	10188	9896	10649
...	...	...	...	...	...
23488	23964	23720	25069	24618	24430
23964	23720	25069	24618	24430	25229
23720	25069	24618	24430	25229	22344
25069	24618	24430	25229	22344	22372
24618	24430	25229	22344	22372	25412

The input is the lagged (prior) values of the variable. The output (forecast horizon) is the next n time steps after the input.

Patterns in Time Series

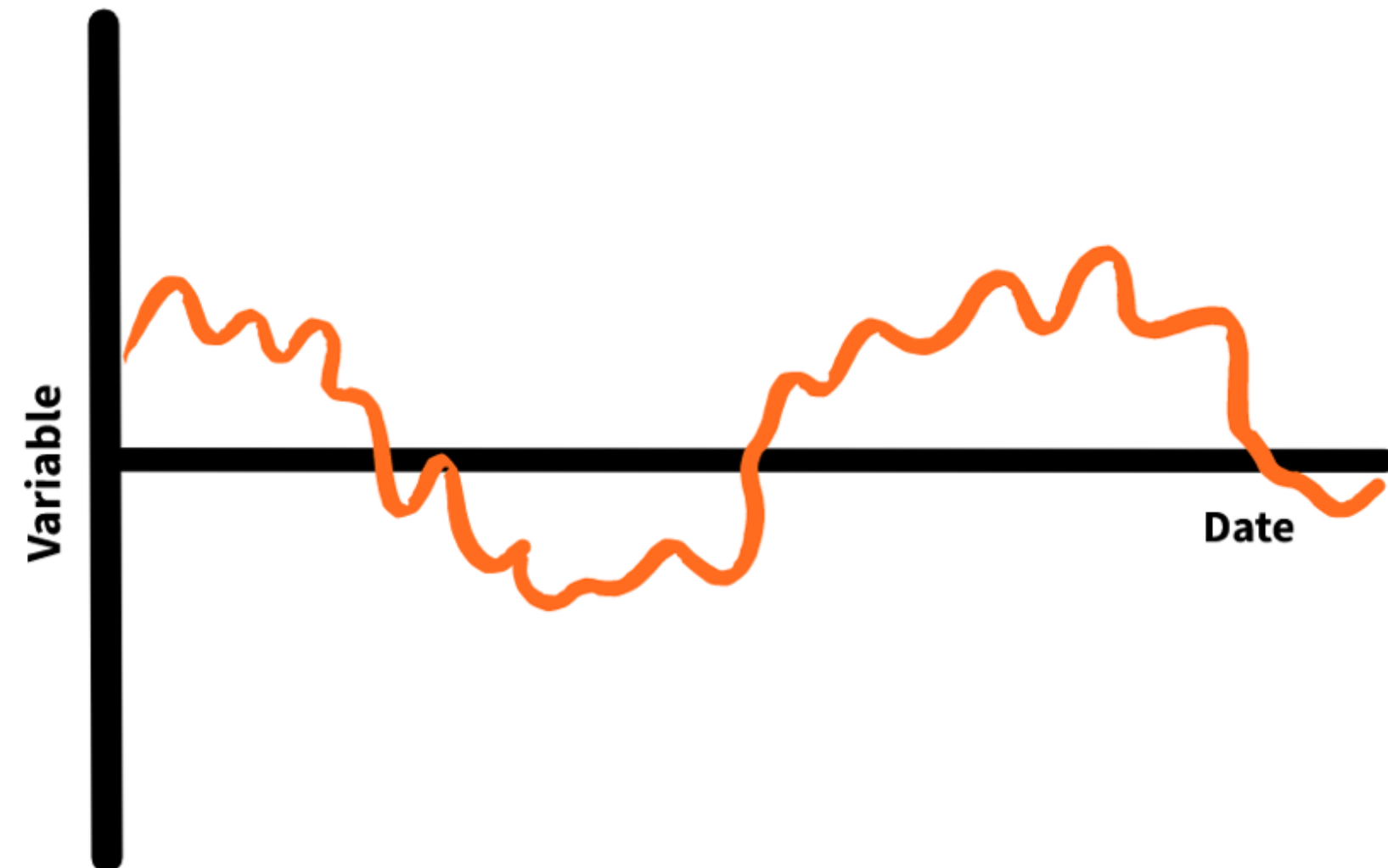
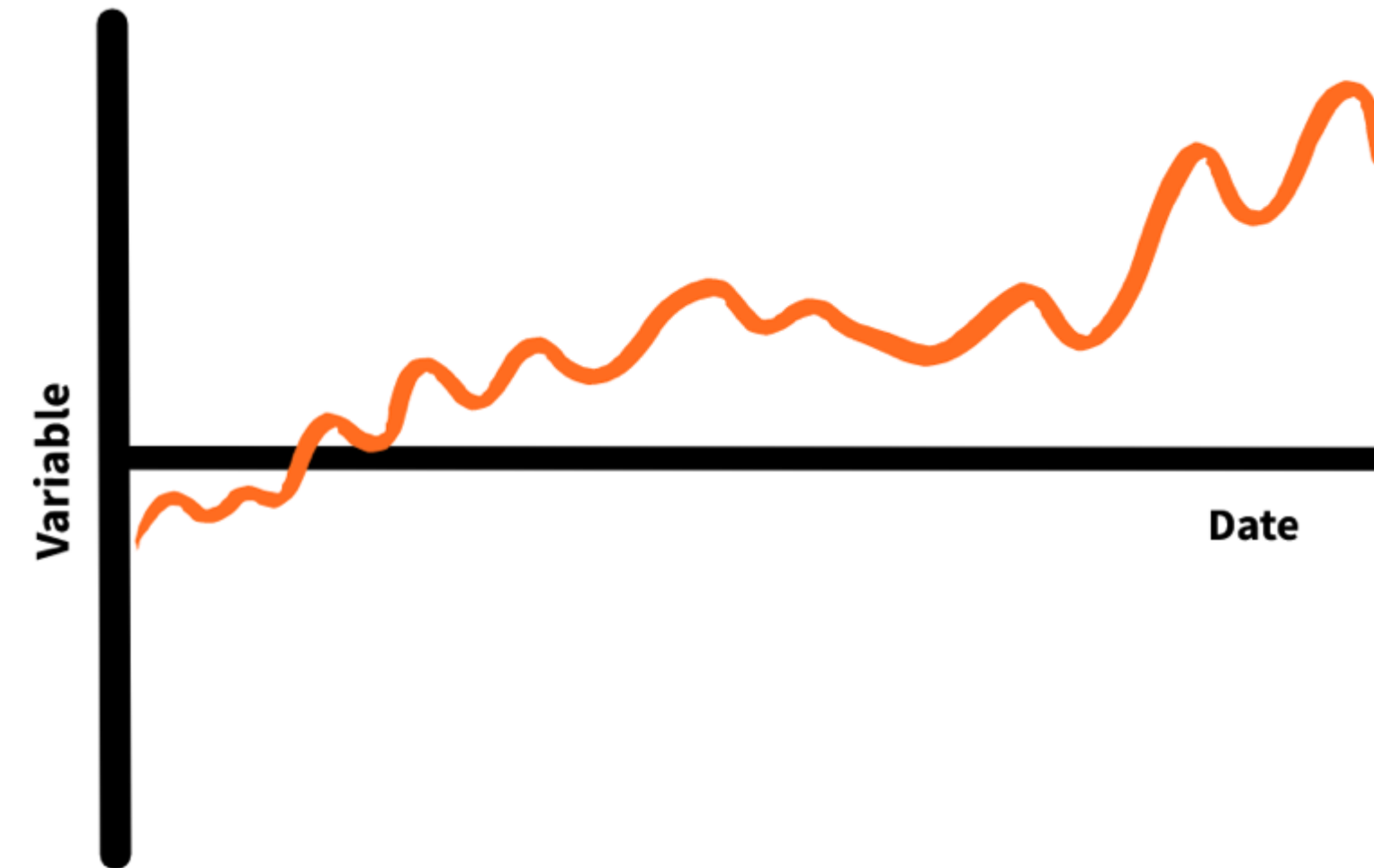
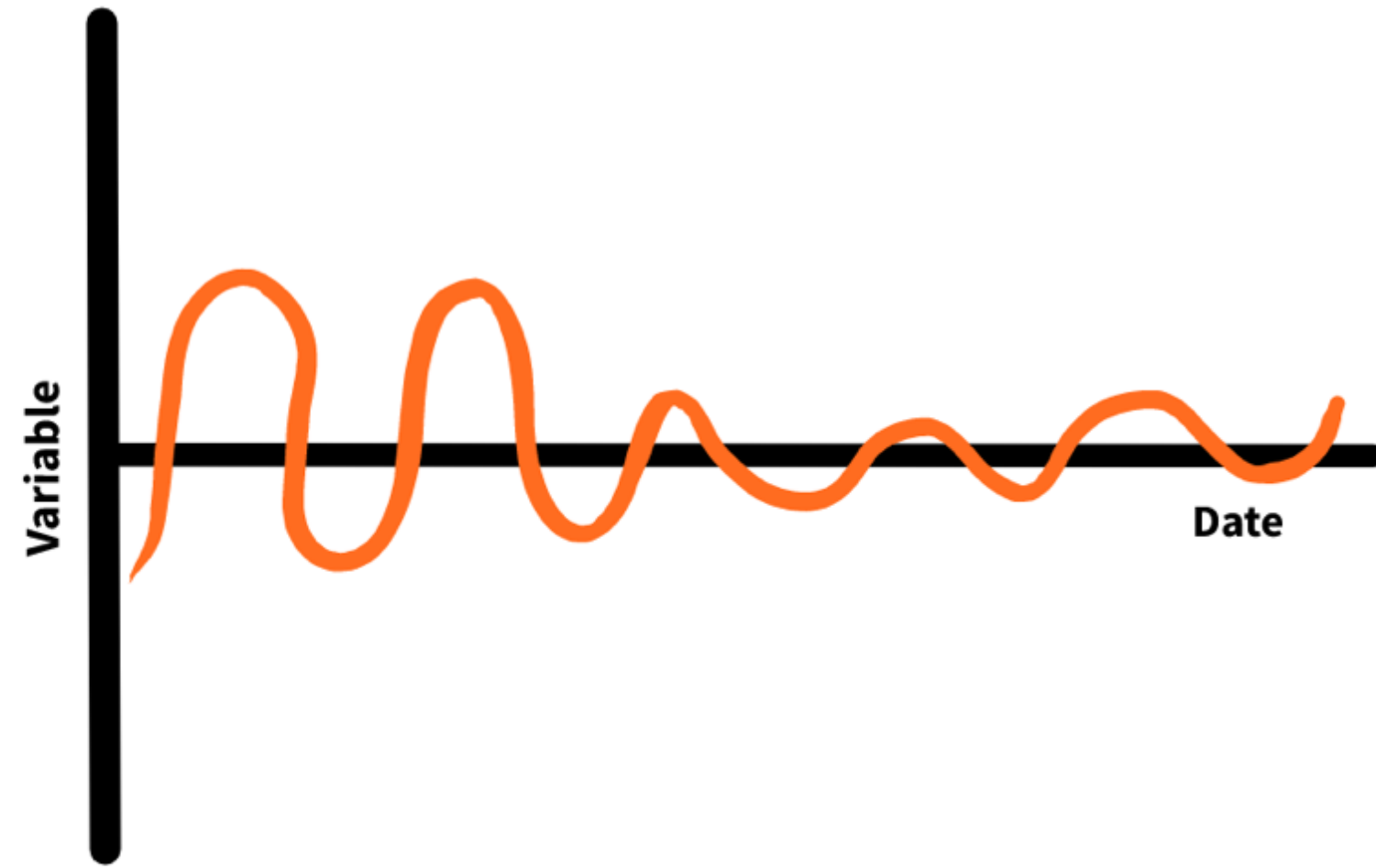
Time series analytics

- Time series analytics should be performed before forecasting!
- Time series analytics is about identifying the intrinsic temporal structure of the time series so we can extrapolate informative patterns to forecast.
- A time series can be decomposed into four components:
 - (1) Trend: the overall increase or decrease of the time series values during a significant time interval
 - (2) Seasonality: periodic fluctuations that recur over a period of less than a year
 - (3) Cyclic patterns: recurring patterns over an unspecified length of time.
 - (4) Noise



Patterns in Time Series

Stationarity

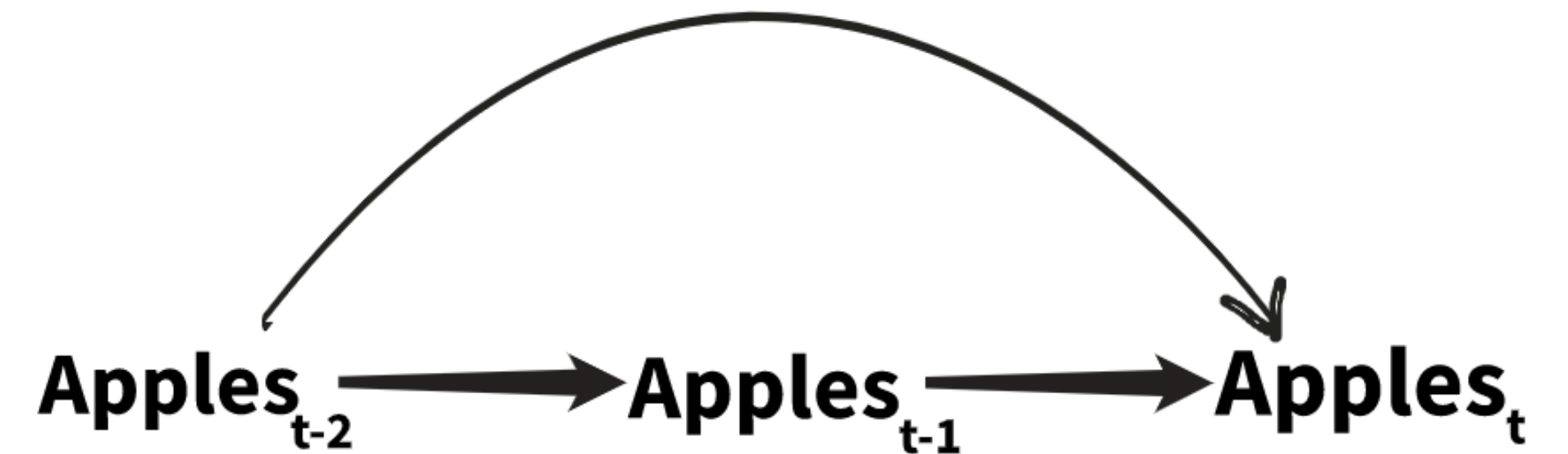


1. Mean is constant
2. Variance is constant
3. No seasonality

Patterns in Time Series

Autocorrelation

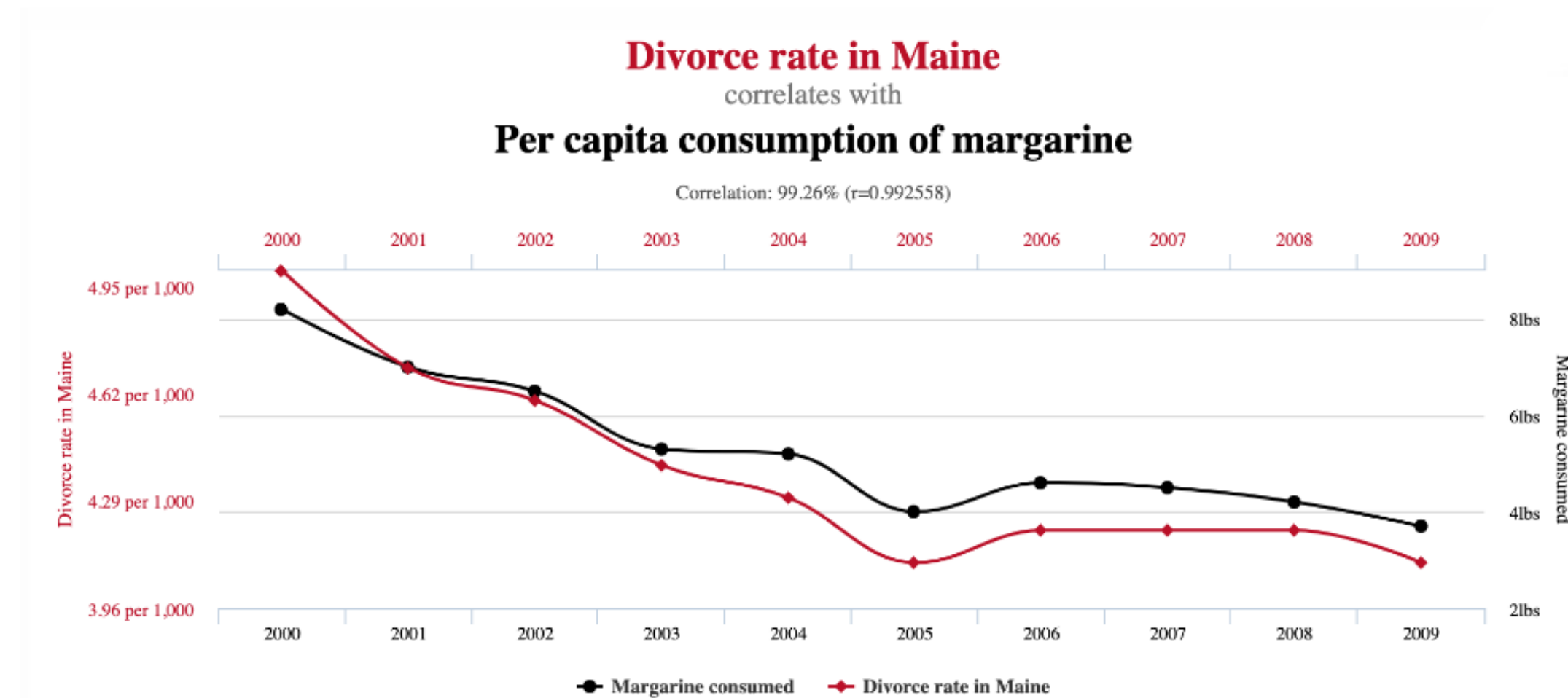
- Autocorrelation is a statistical property that occurs when a time series is linearly related to a previous or lagged version of itself.
- The autocorrelation function (ACF) and partial autocorrelation function (PACF) represent the lagged relationship of a time series



$$\text{ACF} = \text{Apples}_{t-2} + \text{Apples}_t$$

and

$$\text{Apples}_{t-2} + \text{Apples}_{t-1} + \text{Apples}_t$$

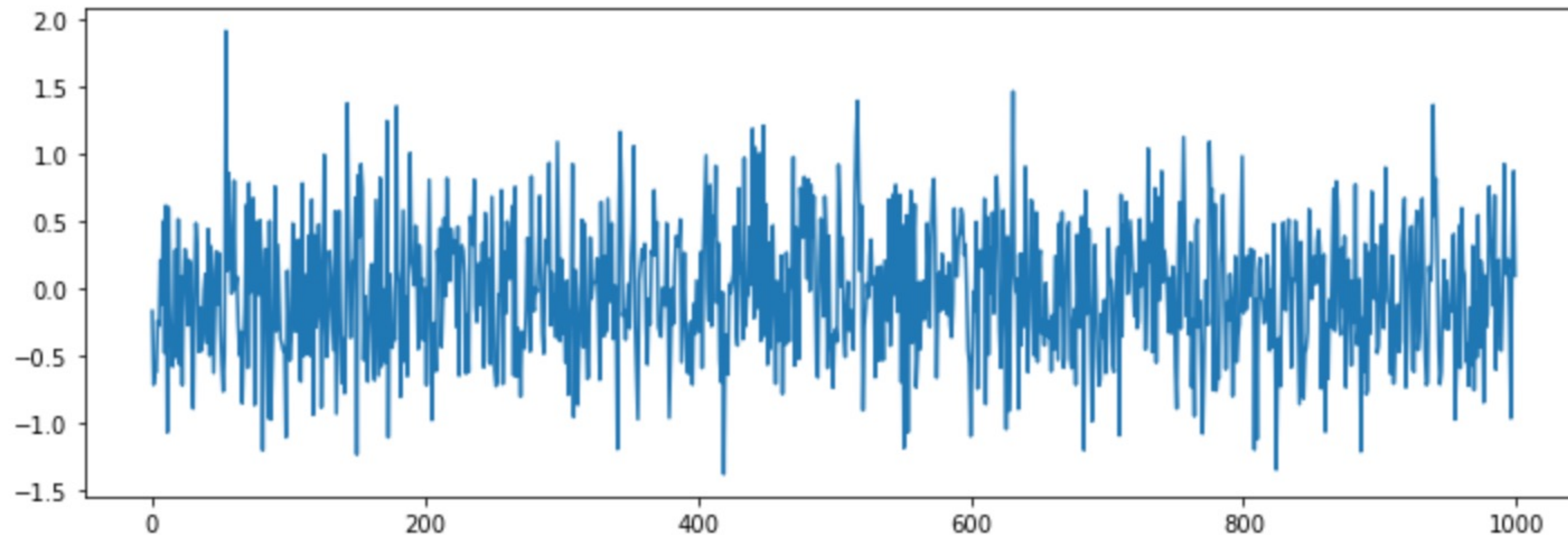
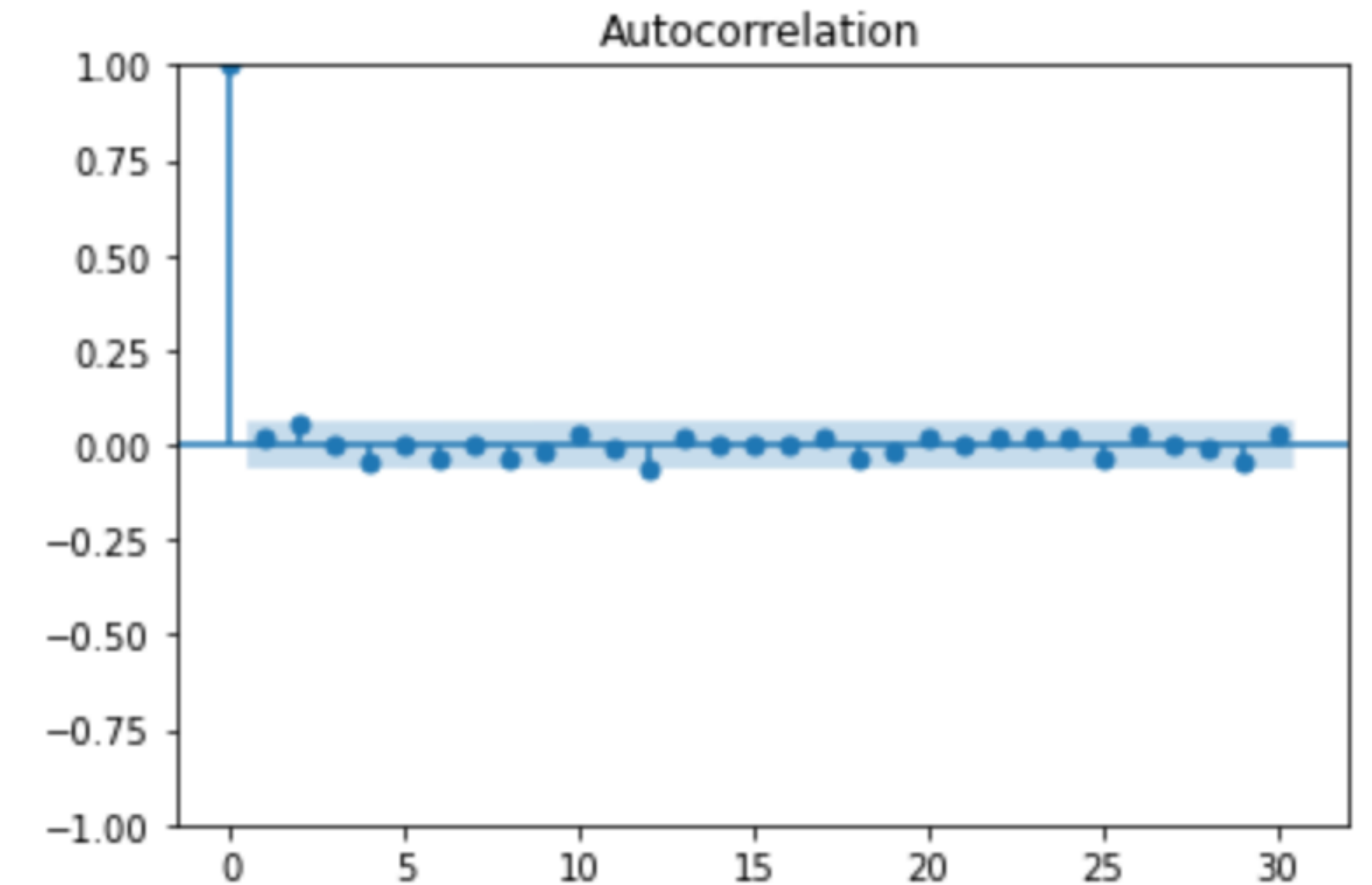


$$\text{PACF} = \text{Apples}_{t-2} + \text{Apples}_t$$

Patterns in Time Series

Autocorrelation of White Noise

- An example of a time series with no autocorrelation is *white noise*
- When the ACF and PACF is plotted there will be no significant relationships



Gaussian White Noise

An Example of Time Series Forecasting with ARIMA

Classical Forecasting Techniques

Autoregressive Integrated Moving Average (ARIMA)

- Composed of AutoRegression (AR), Differencing (or Integration, I), and Moving Average (MA).
- ARIMA assumes that future values can be represented as a linear function of differenced observations and residual errors at prior (lagged) time steps.
- Each part of ARIMA has a parameter associated with it. A standard notation is $ARIMA(p, d, q)$, where the parameters are substituted with integer values to specify the specific ARIMA model.
- The parameter p can be determined with a PACF. The parameter q can be determined with the ACF.

Classical Forecasting Techniques

Seasonal Autoregressive Integrated Moving Average (SARIMA)

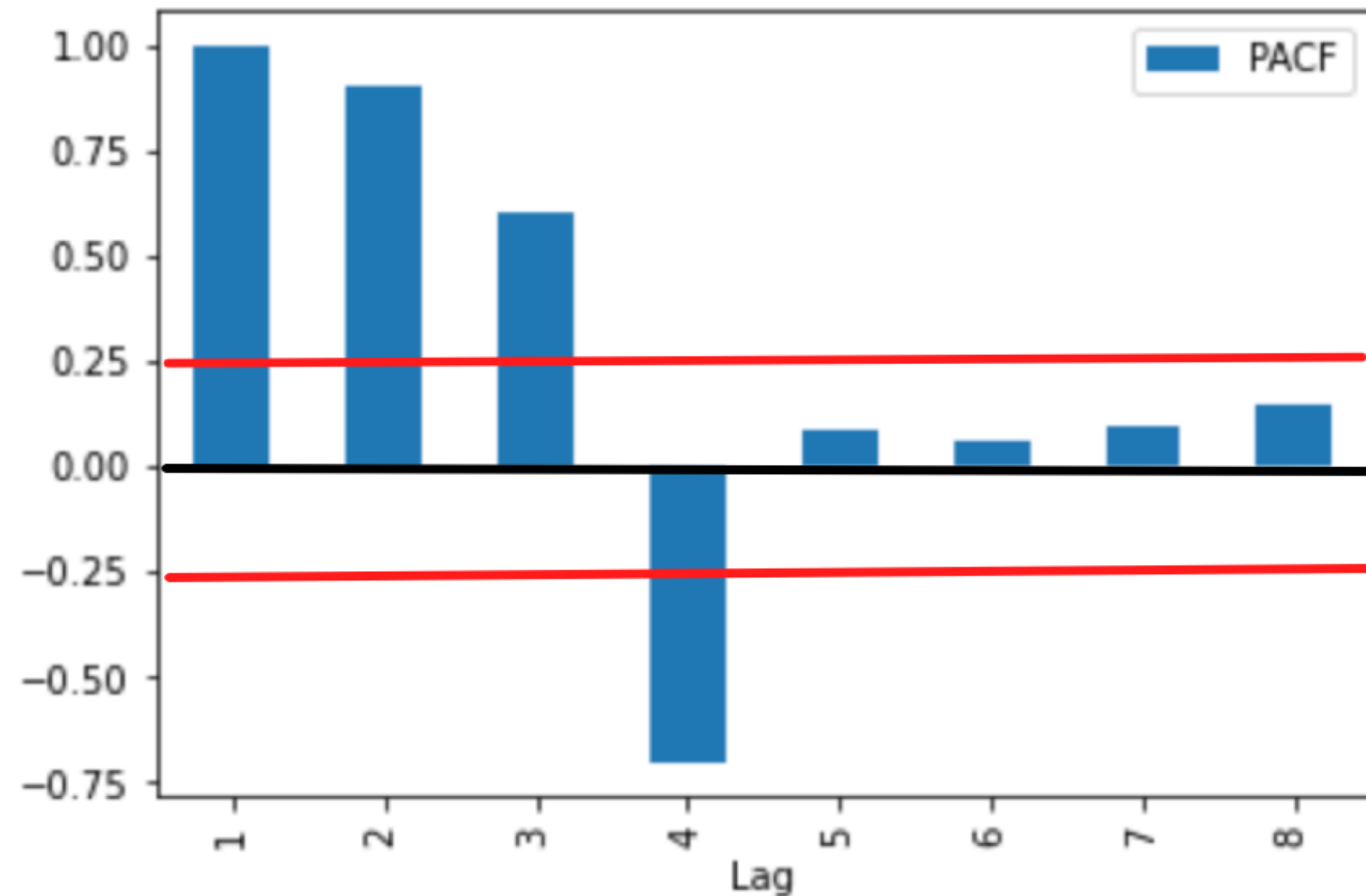
- A problem with ARIMA is that it does not account for seasonality. Seasonal ARIMA or SARIMA is an extension of ARIMA that explicitly supports seasonality.
- Four additional parameters are included with SARIMA compared to ARIMA:
 - P : seasonal autoregressive order
 - D : seasonal difference order
 - Q : the seasonal moving average order
 - m : the number of time steps for a seasonal period
- This results in the following notation: $SARIMA(p, d, q)(P, D, Q)_m$

Importantly, the m parameter influences the P , D , and Q parameters. For example, an m of 12 for monthly data suggests a yearly seasonal cycle. A $P = 1$ would use the first seasonally offset observation in the model.

Classical Forecasting Techniques

Determining parameters of SARIMA

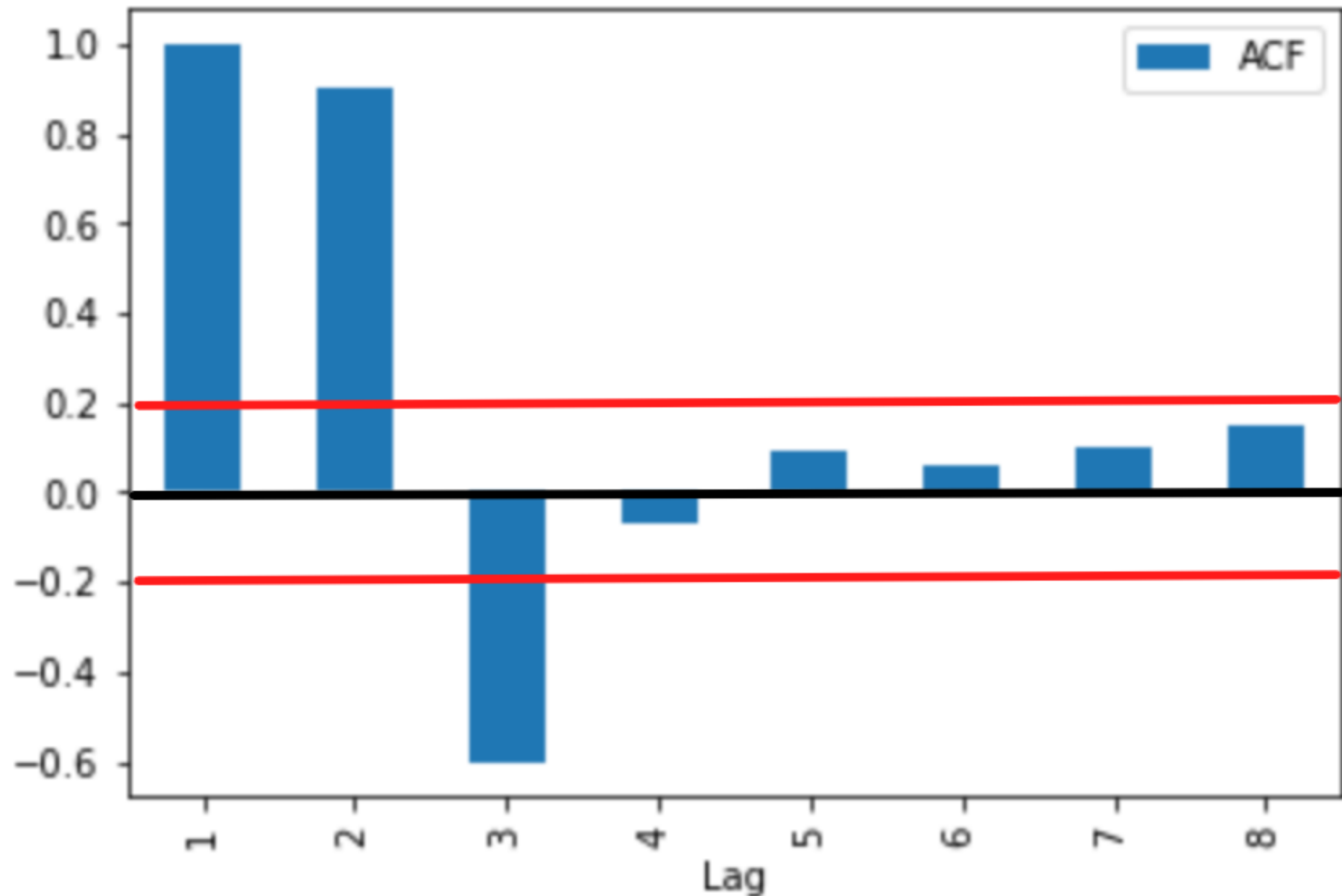
- We can determine the AR or Autoregressive term, p , with the PACF. In this toy example, p would be 4.



Classical Forecasting Techniques

Determining parameters of SARIMA

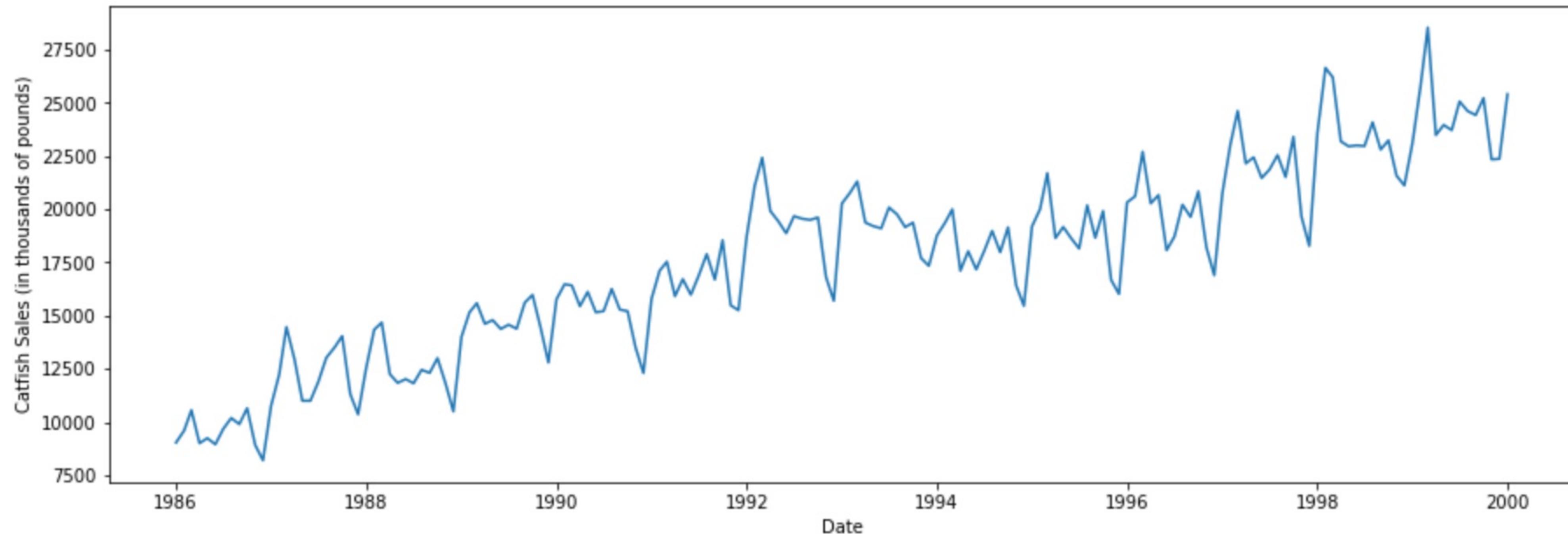
- Similarly, we can determine the MA or Moving Average term, q , with the ACF. In this toy example, p would be 3.



Classical Forecasting Techniques

Time Series Forecasting Example with SARIMA

Goal: Forecast catfish sales time series

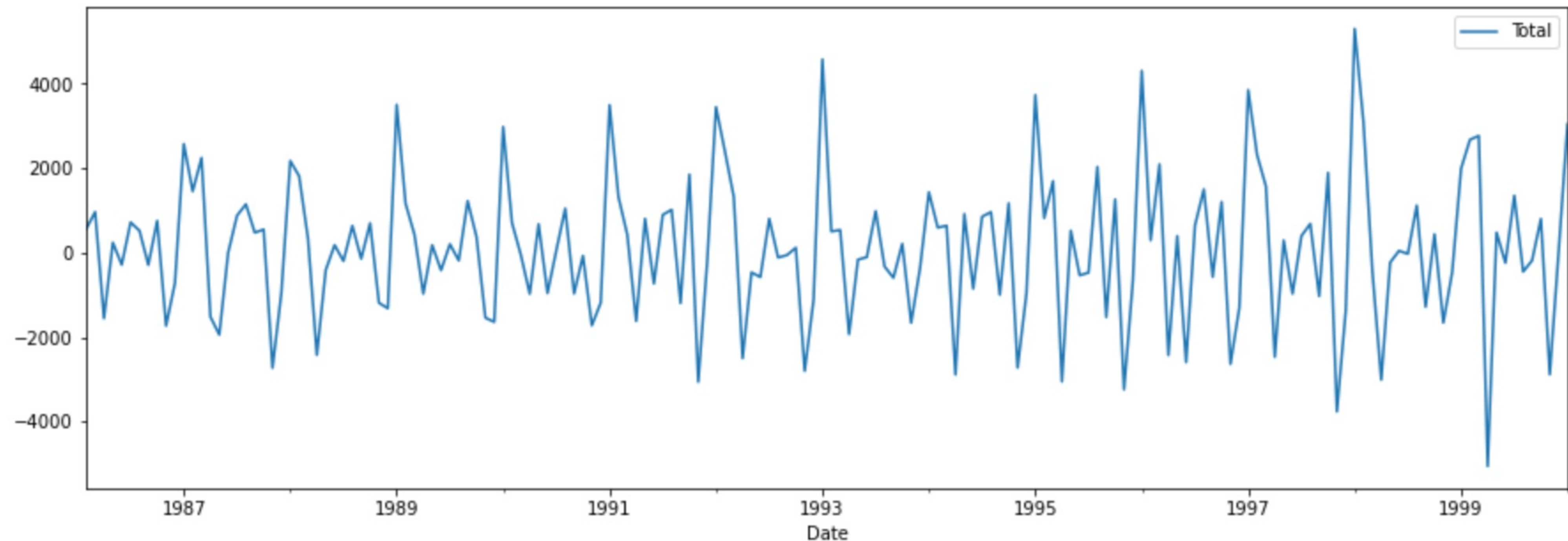


Classical Forecasting Techniques

Time Series Forecasting Example with SARIMA

First, let's difference the time series to eliminate the trend. Differencing is computed by subtracting the current value subtracted by the past value, or:

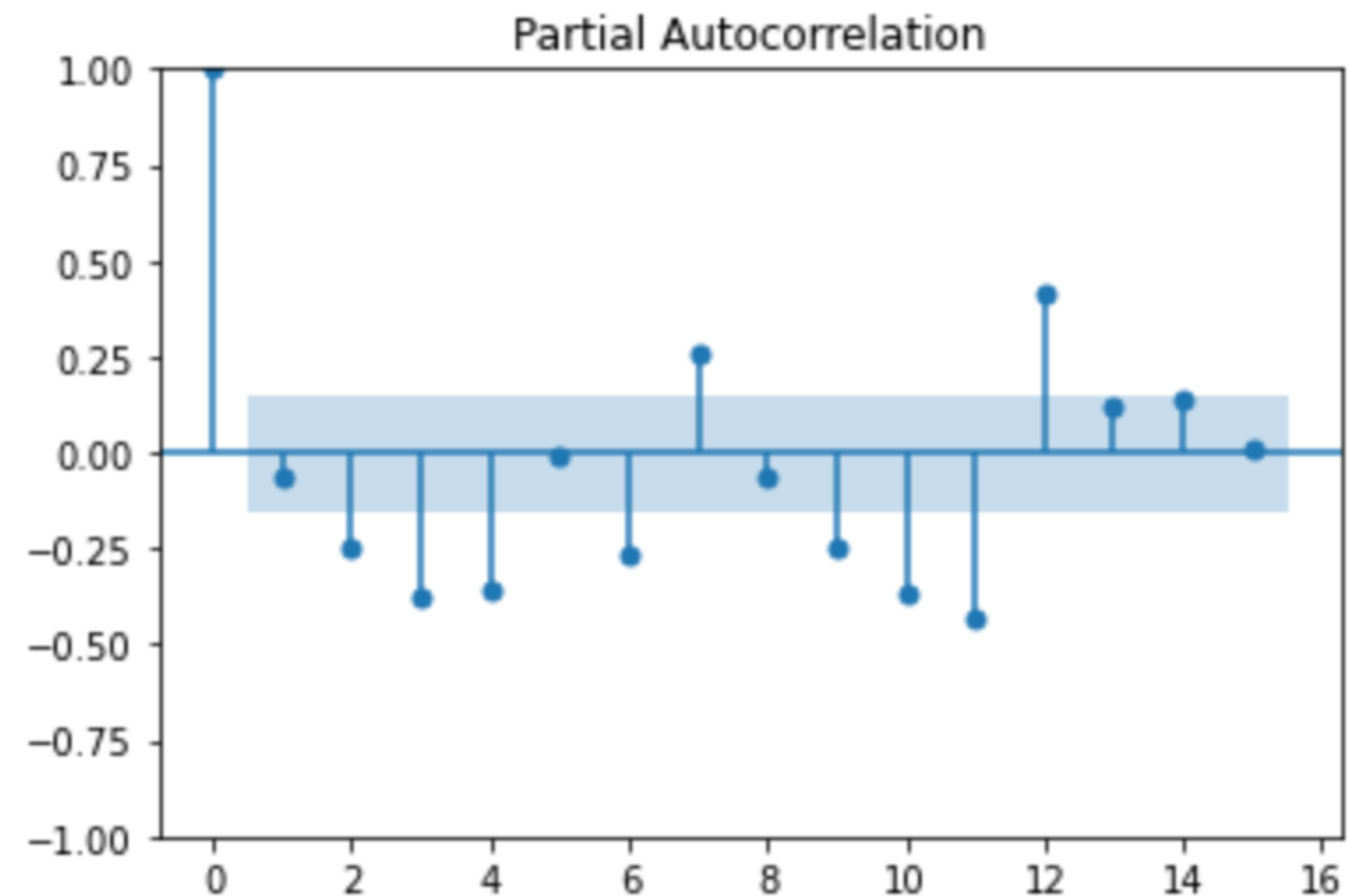
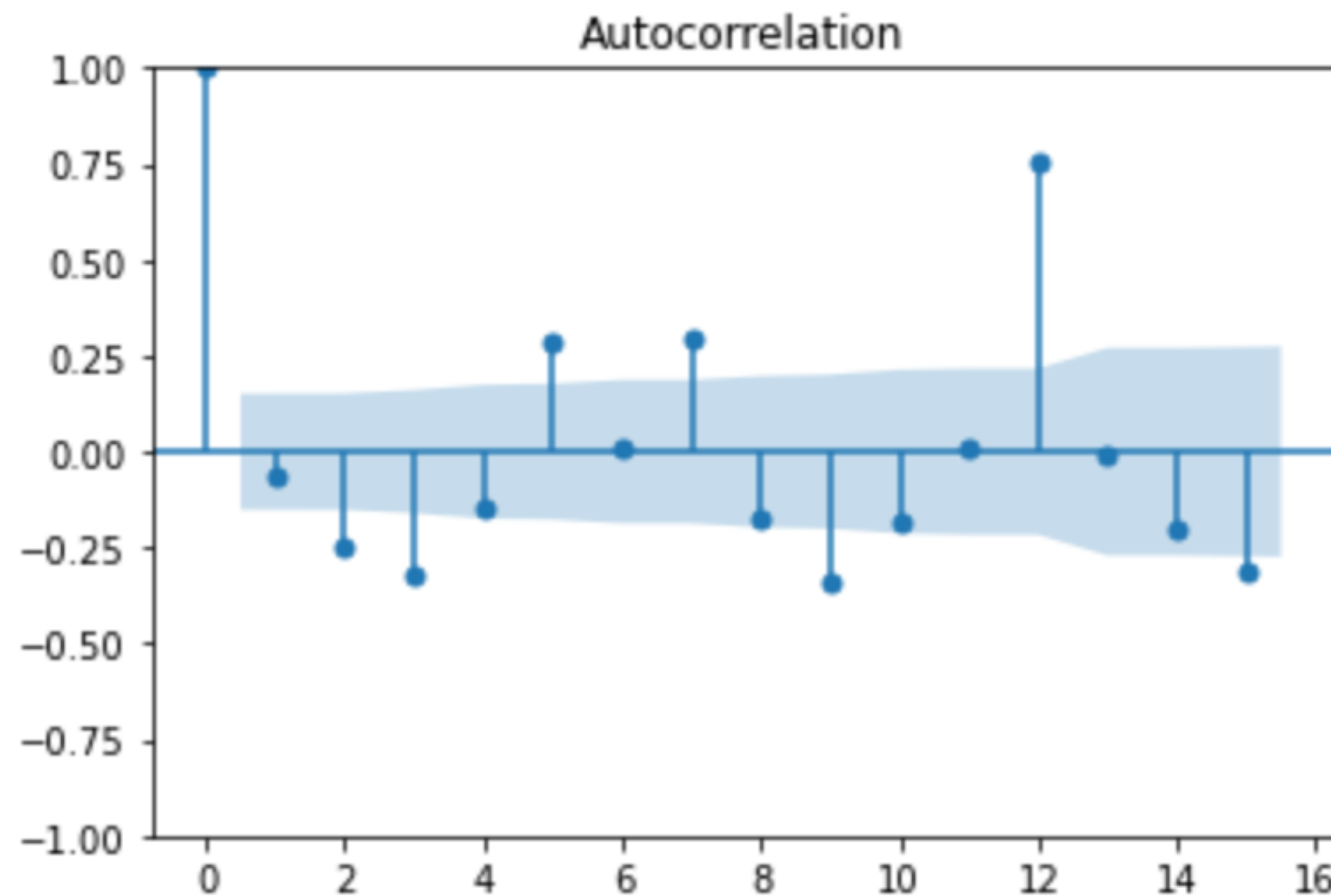
$$y_t = y_t - y_{t-1}$$



Classical Forecasting Techniques

Time Series Forecasting Example with SARIMA

Next, let's find the p and q terms with the ACF and PACF plots. Based on the preceding ACF and PACF plots, we will try a seasonal (12 months) AR and MA process.



Classical Forecasting Techniques

Time Series Forecasting Example with SARIMA

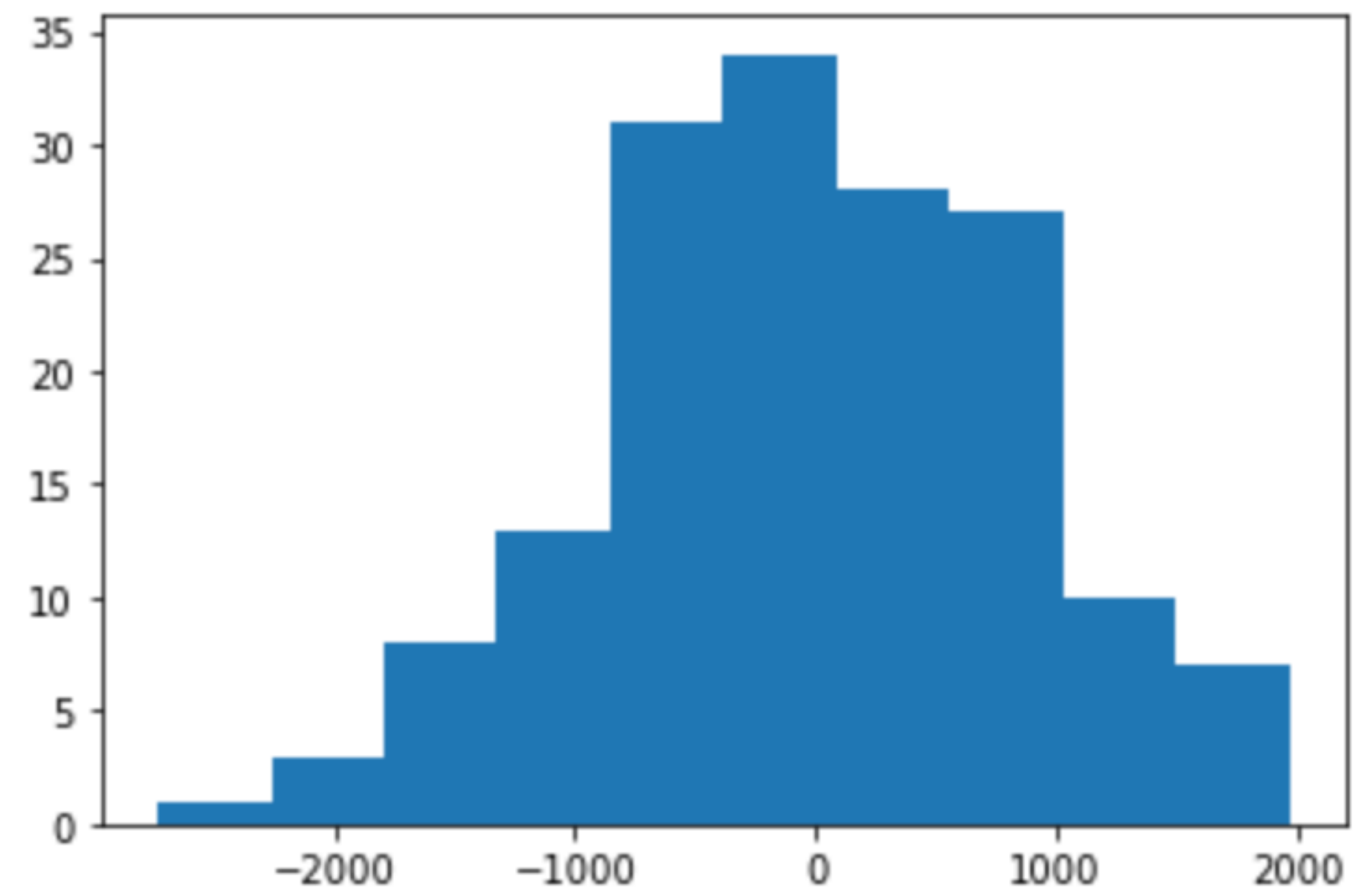
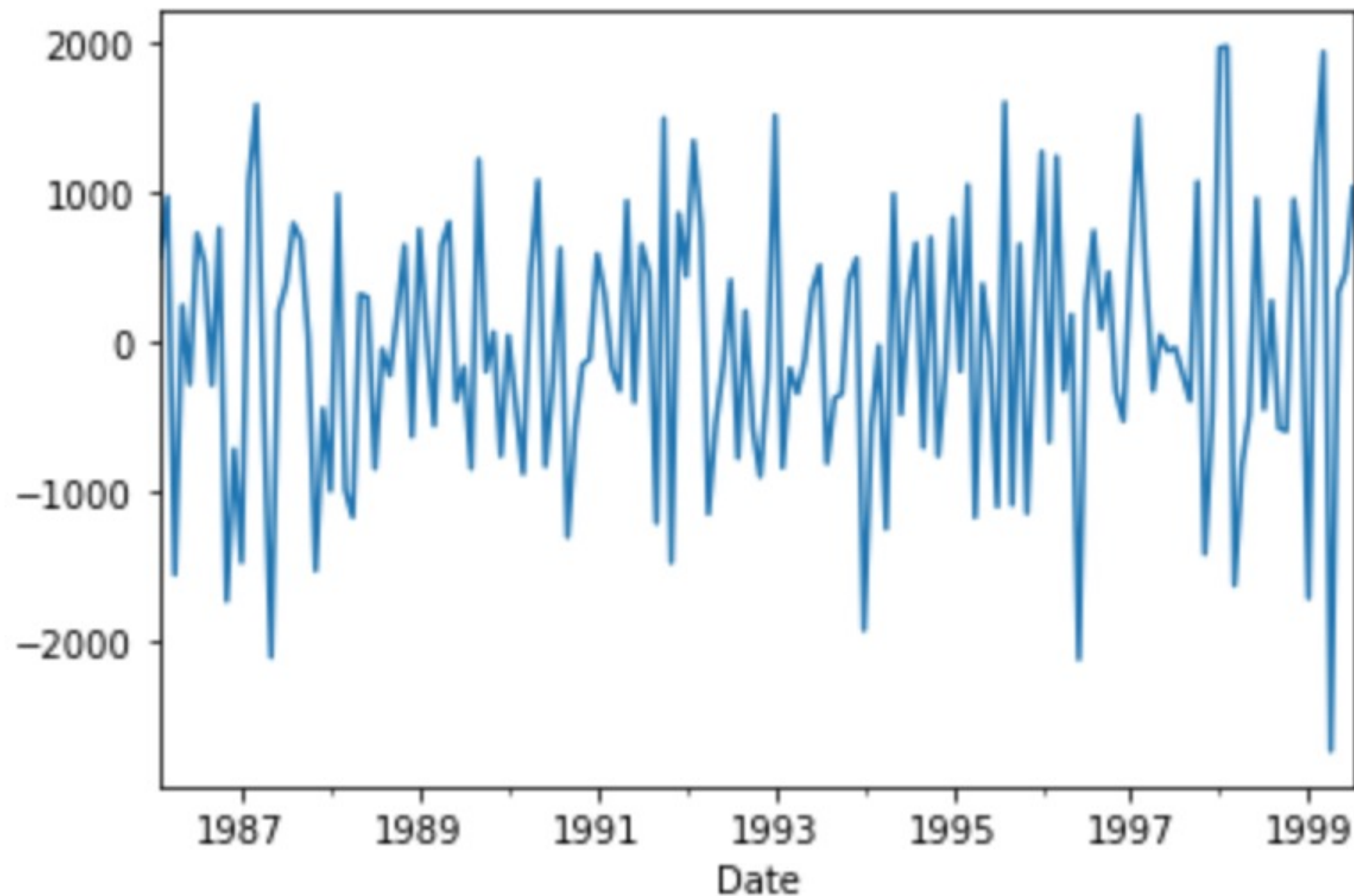
- Since we differenced our time series once the d parameter will be 1, the seasonal P and Q terms are 1 with an m of 12 to account for the annual seasonality. This results in a $SARIMA(0,1,0)(1,0,1)_{12}$ model.
- We train the SARIMA model on data from 1986-01-01 to 1999-07-01. The following summary is provided to describe the fitted model:

SARIMAX Results						
Dep. Variable:		Total		No. Observations:		163
Model:		SARIMAX(0, 1, 0)x(1, 0, [1], 12)		Log Likelihood		-1329.806
Date:		Mon, 21 Mar 2022		AIC		2665.611
Time:		19:09:22		BIC		2674.874
Sample:		01-01-1986		HQIC		2669.372
		- 07-01-1999				
Covariance Type:		opg				
	coef	std err	z	P> z	[0.025	0.975]
ar.S.L12	0.9873	0.012	84.116	0.000	0.964	1.010
ma.S.L12	-0.7829	0.089	-8.778	0.000	-0.958	-0.608
sigma2	6.874e+05	7.75e+04	8.866	0.000	5.35e+05	8.39e+05
Ljung-Box (L1) (Q):		8.53	Jarque-Bera (JB):		0.62	
Prob(Q):		0.00	Prob(JB):		0.73	
Heteroskedasticity (H):		2.19	Skew:		-0.14	
Prob(H) (two-sided):		0.00	Kurtosis:		3.10	

Classical Forecasting Techniques

Time Series Forecasting Example with SARIMA

We check the residuals (fitted values minus the actual values) to ensure that there is only white noise. If not, we have missed some of the signal from the time series. Based on the residuals plotted below, we can be confident that we capture the signal and that the parameters we choose are appropriate.



Classical Forecasting Techniques

Time Series Forecasting Example with SARIMA

We are ready to forecast, but how do we evaluate it? Many metrics exist, but we will use the Mean Average Percentage Error or MAPE which is defined:

$$M = \frac{1}{n} \sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right|$$

M = mean absolute percentage error

n = number of times the summation iteration happens

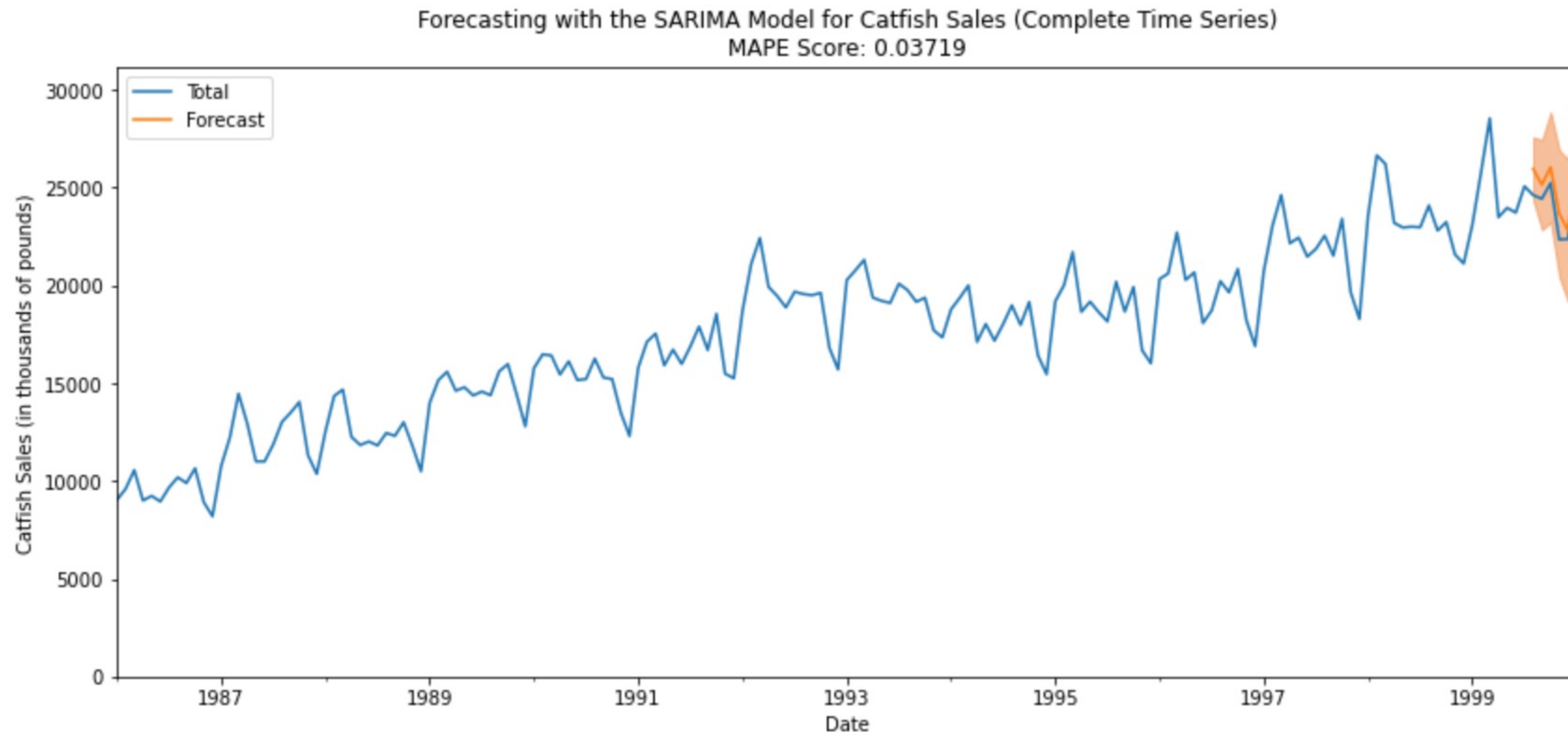
A_t = actual value

F_t = forecast value

Classical Forecasting Techniques

Time Series Forecasting Example with SARIMA

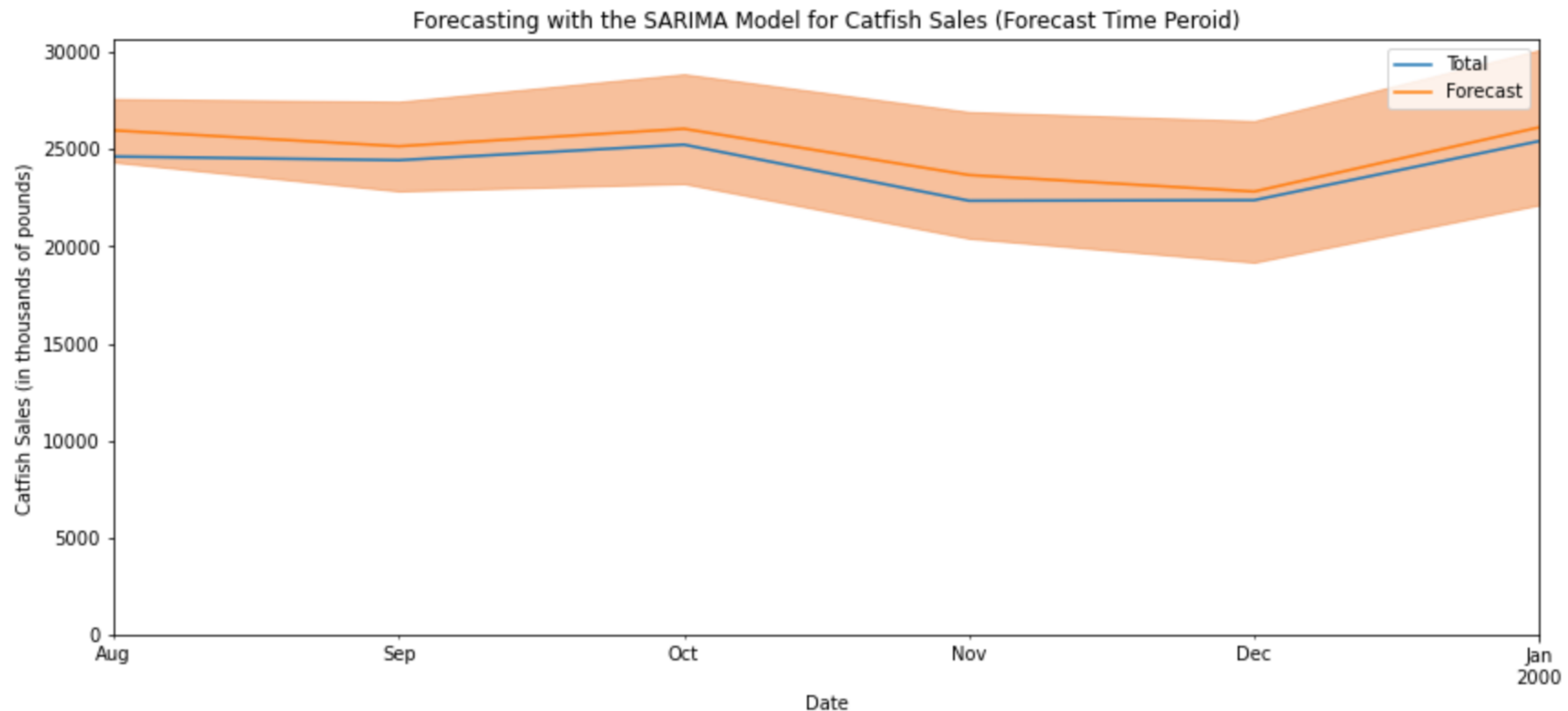
Now we are finally ready to forecast! We produce a 6-step forecast, with a 95% prediction interval.



Classical Forecasting Techniques

Time Series Forecasting Example with SARIMA

Now we are finally ready to forecast! We produce a 6-step forecast, with a 95% prediction interval.



Hybrid Deep Learning for Time Series Forecasting

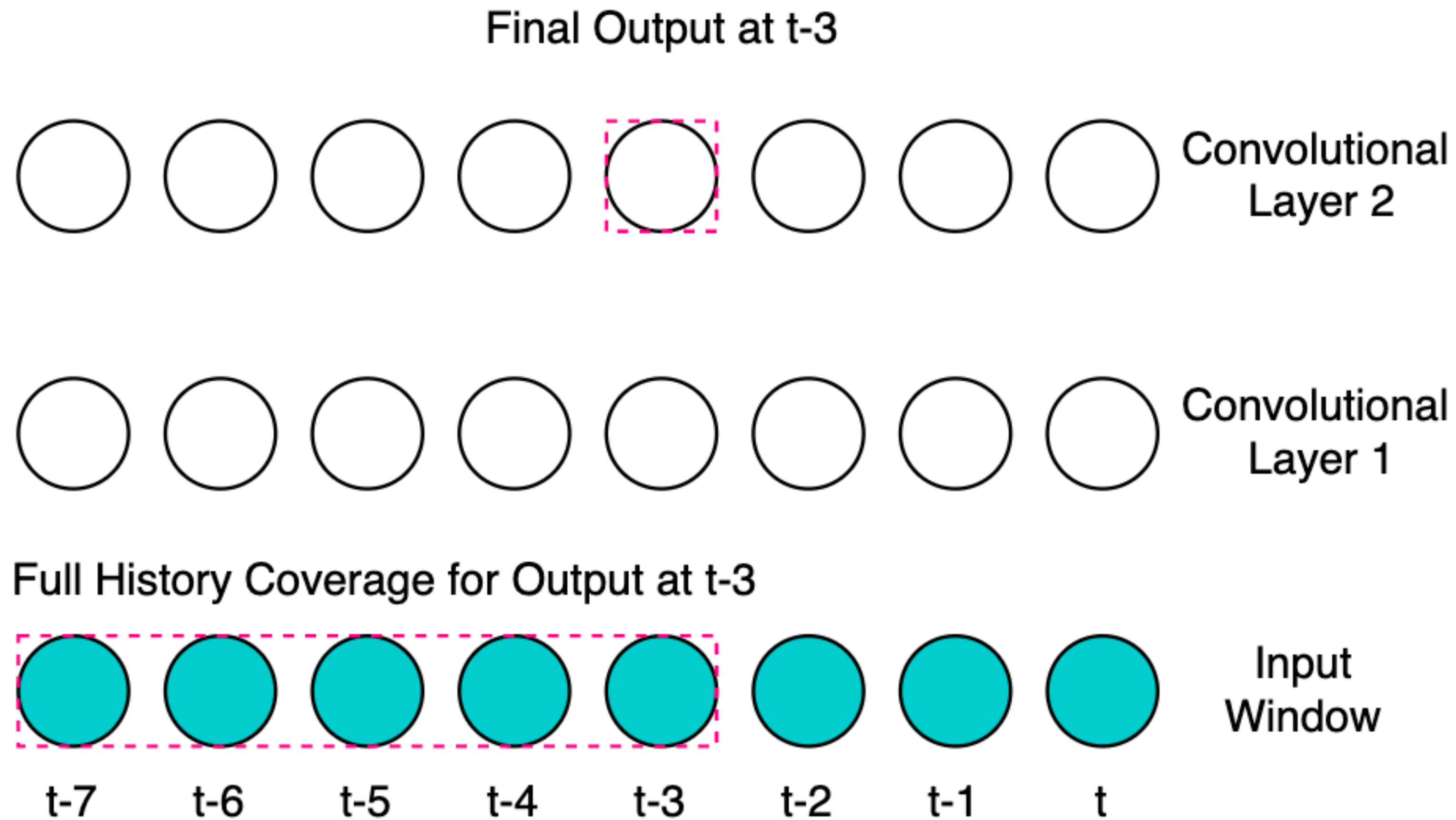
Issues in Health Forecasting

Limitations of past research in health forecasting with hybrid methods

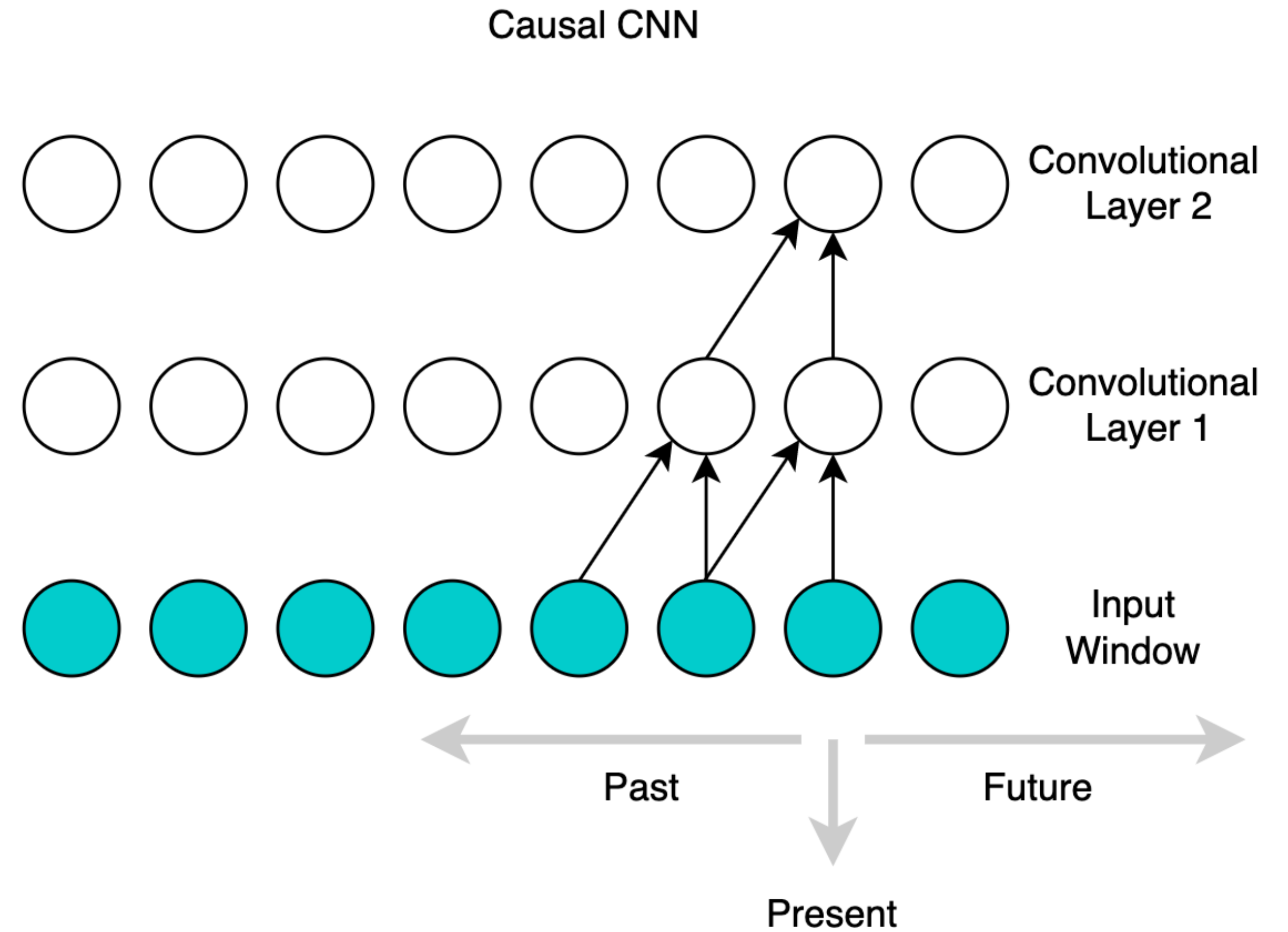
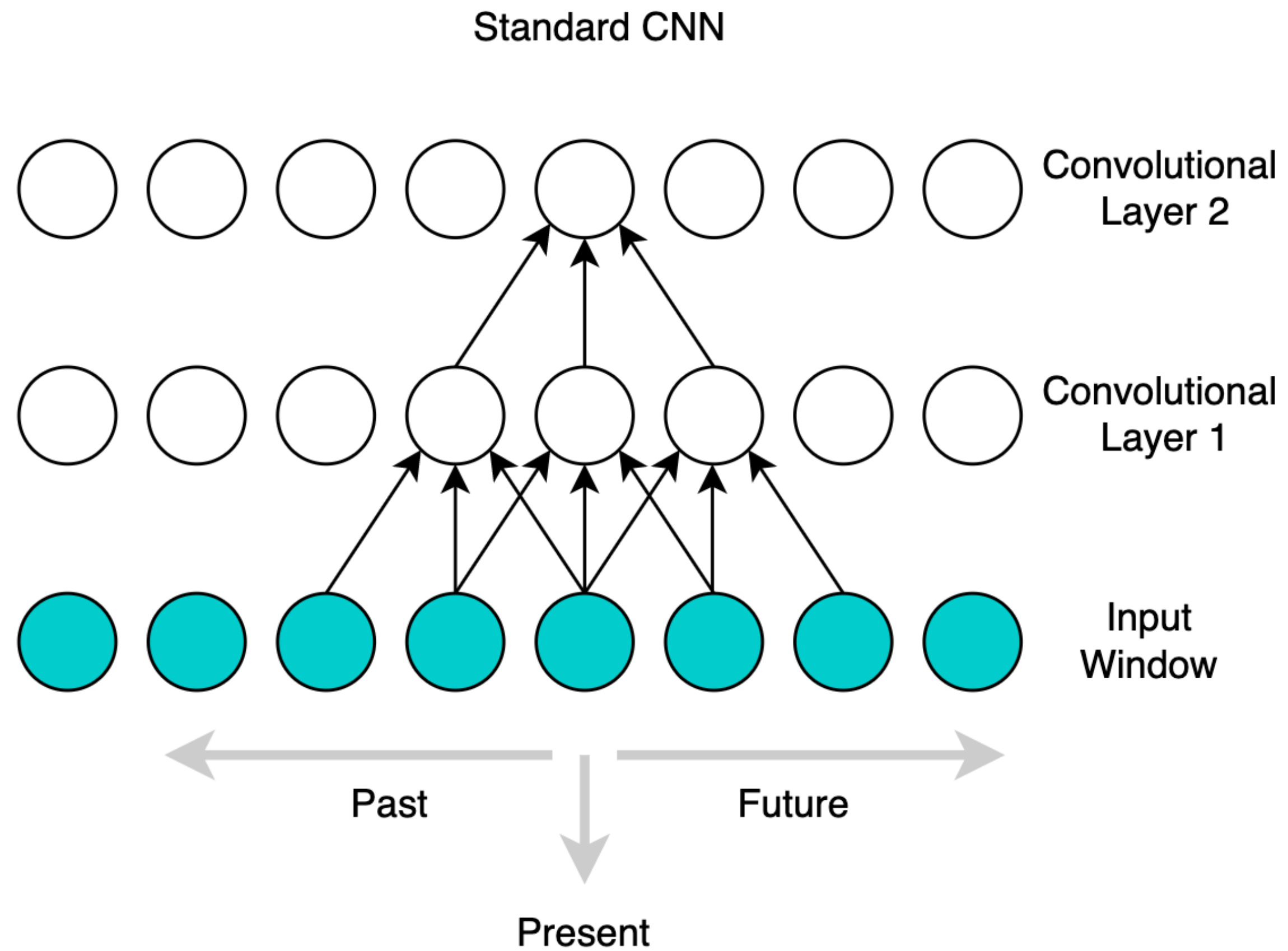
Limited research has been conducted with the following approaches:

1. Fully history coverage
2. Causal convolutions
3. Multivariate time series data

1. Full History Coverage



2. Causal Convolutions



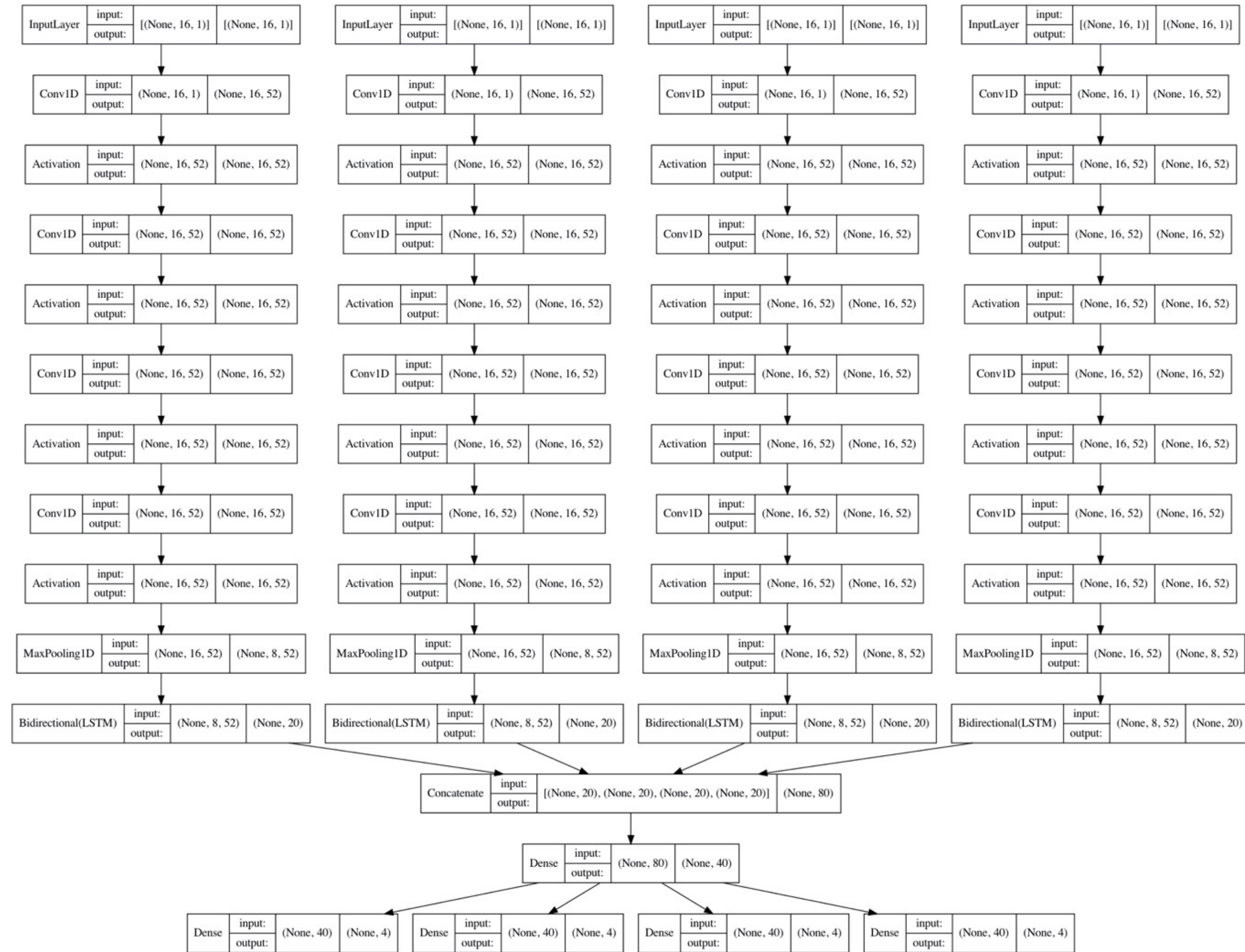
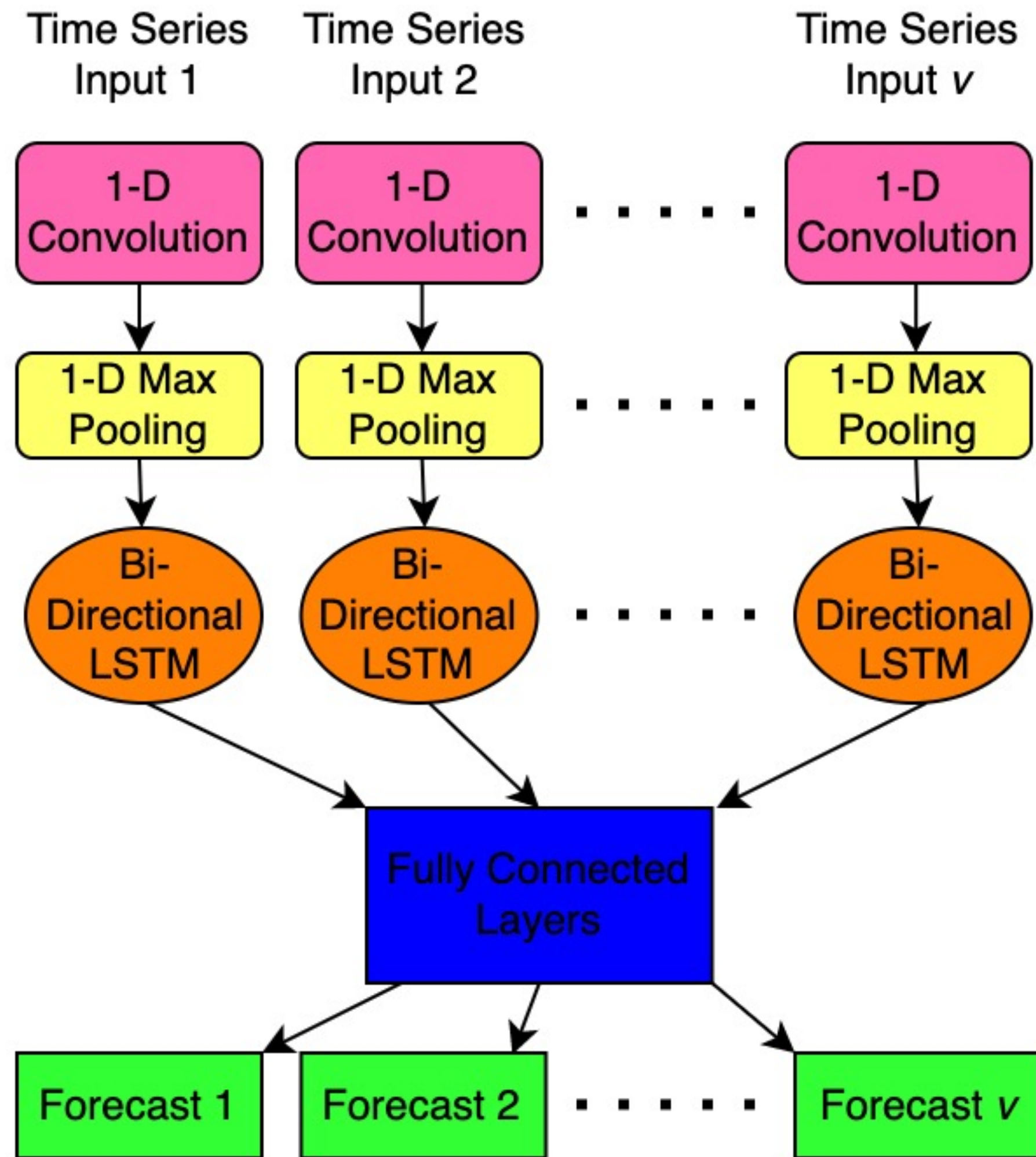
3. Multivariate

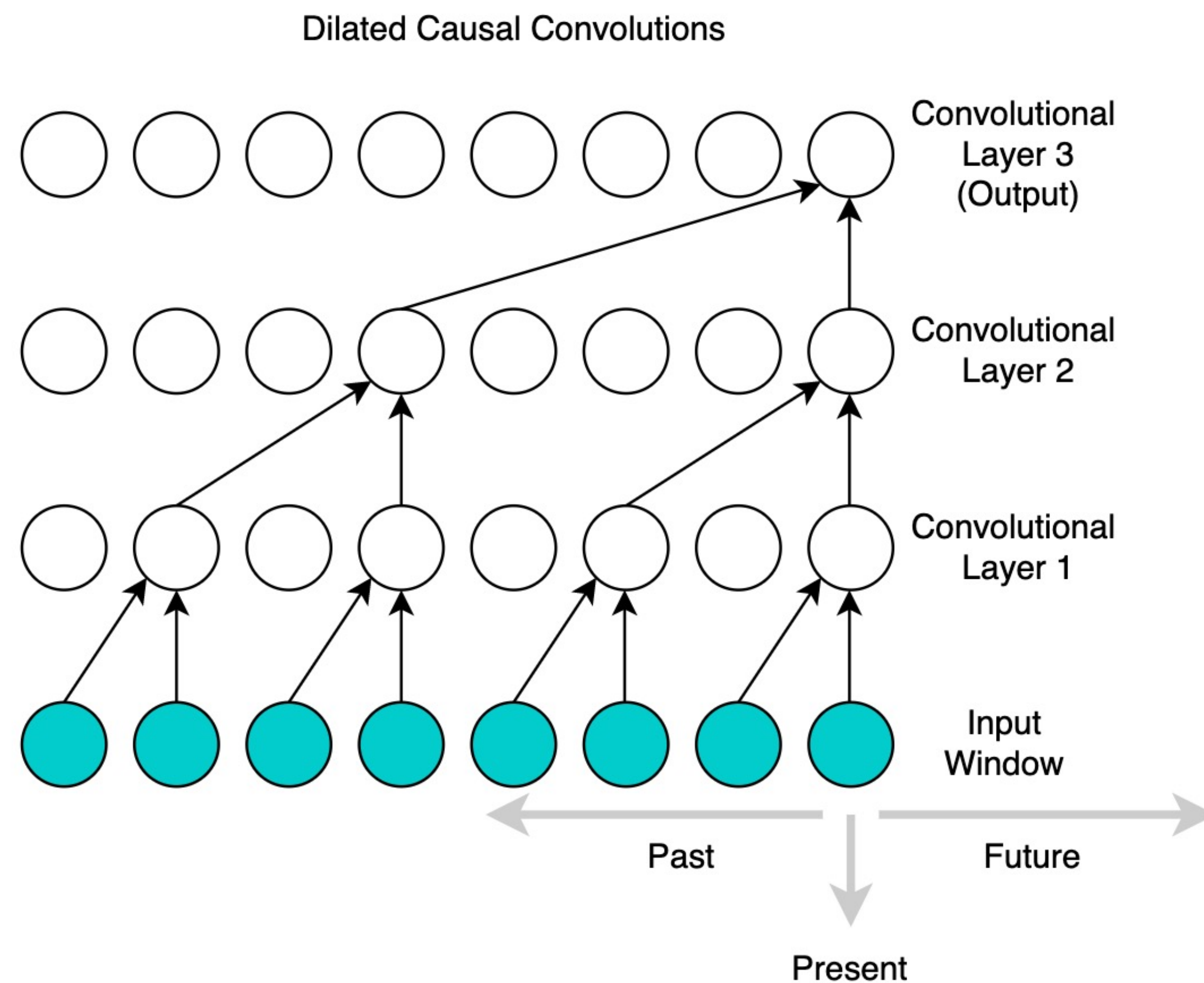
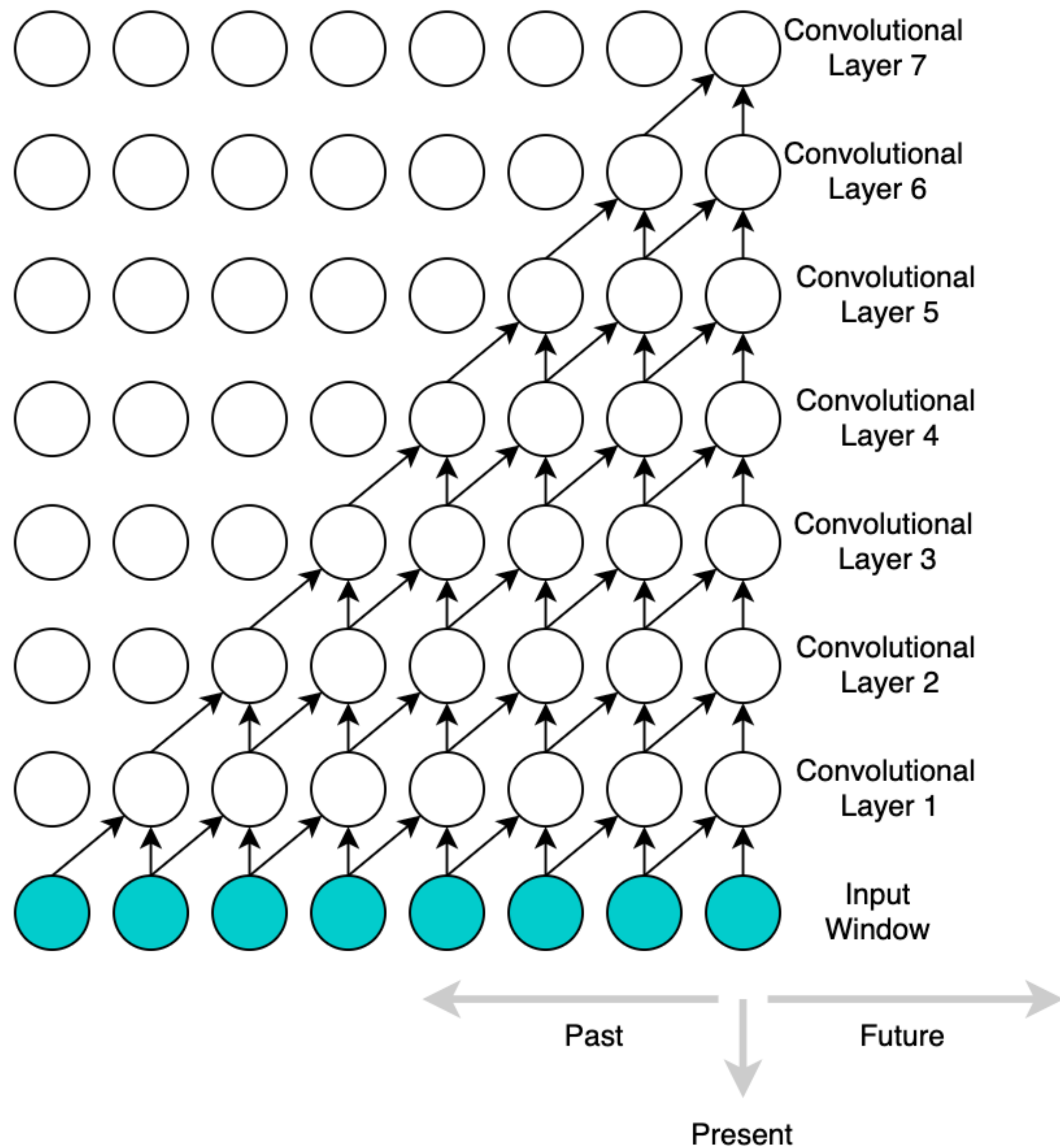
- Most studies involve a single time series (univariate).
- Multivariate time series can exploit the relationships between multiple input variables and the output.
- Additionally, the relationships between the input variables can be used to increase forecasting accuracy.

The proposed SCNN-BiLSTM architecture for health forecasting

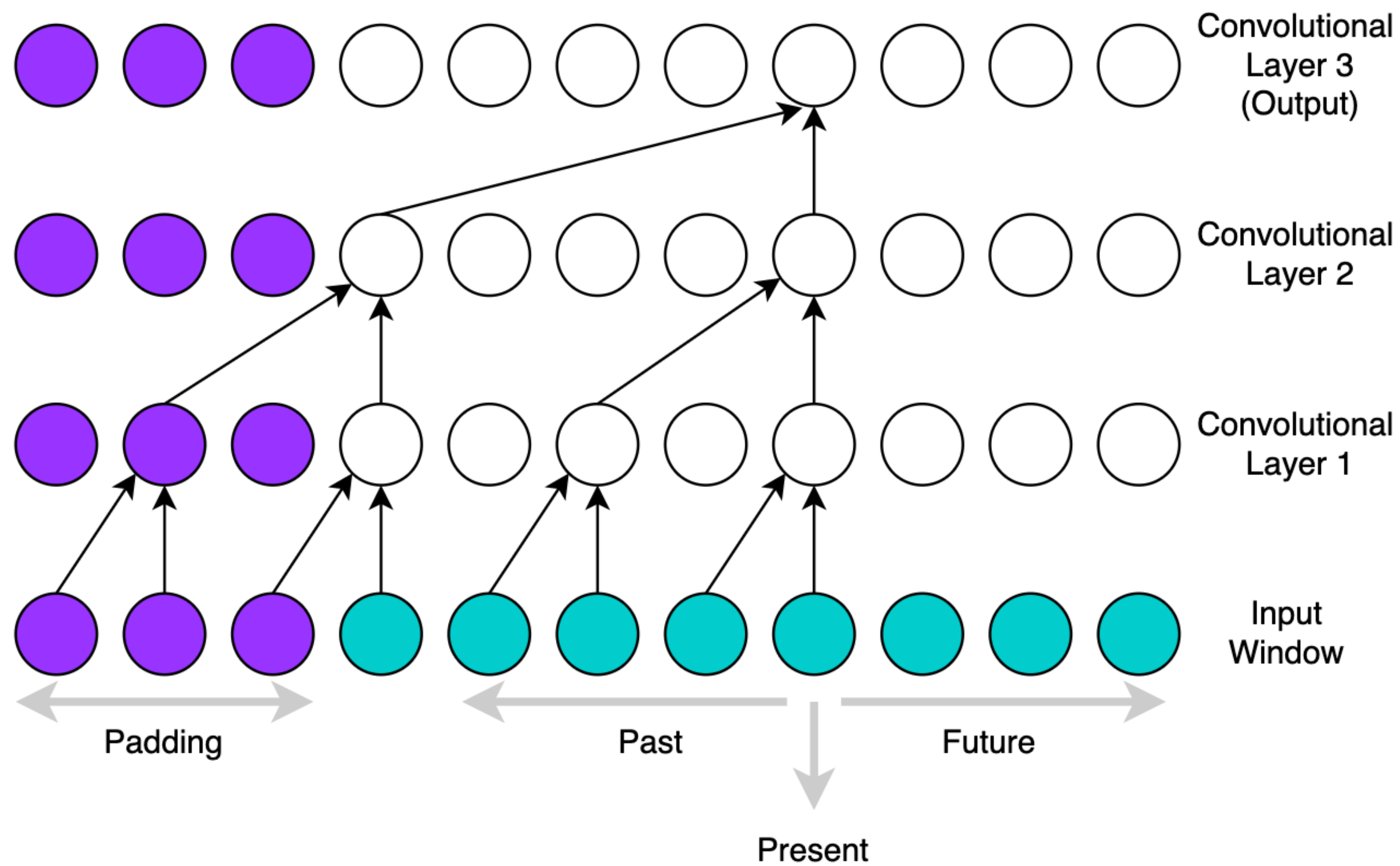
The proposed hybrid architecture extends previous research in the following ways:

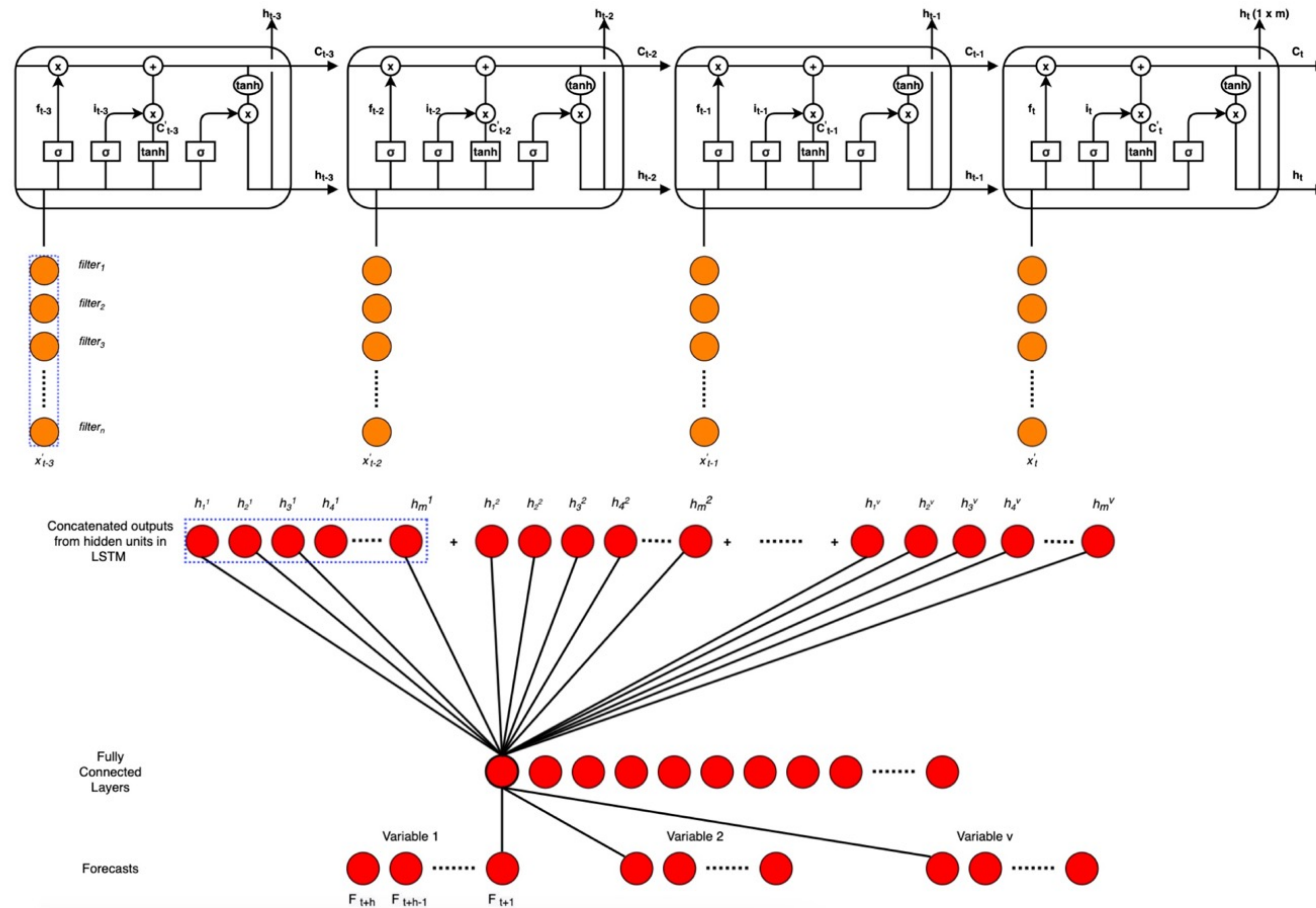
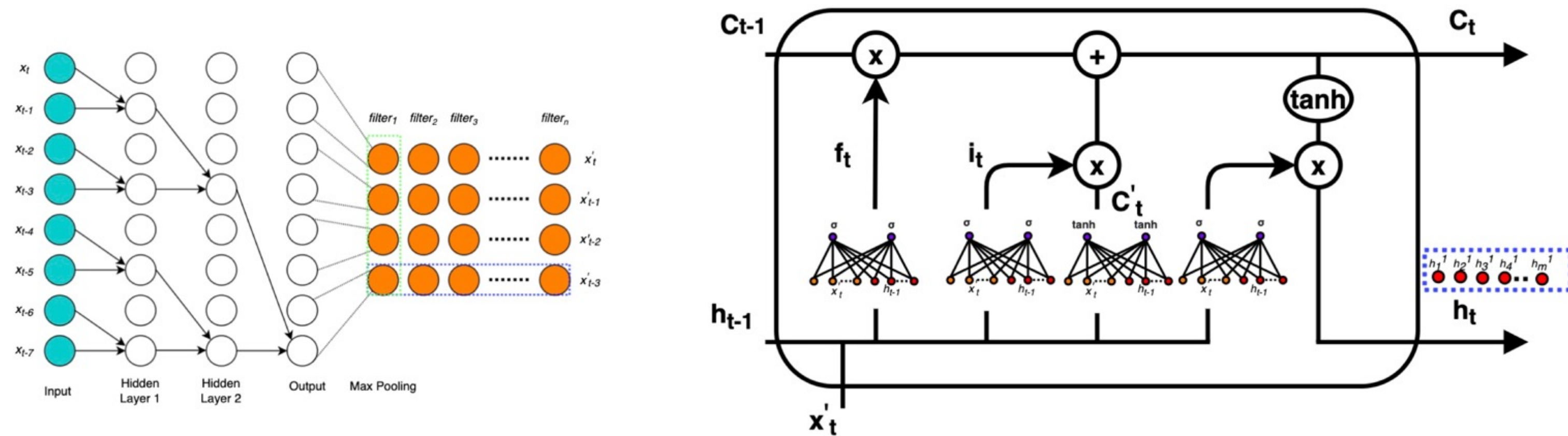
1. Stacked-dilated-causal CNN layers provide full history coverage that maintains the causal structure of the time series.
2. Incorporation of multivariate data.
3. Automatic inference of incomplete multivariate data.
4. Multi-headed architecture, so a separate CNN and RNN process each time series input.
5. Longer-range forecast horizon.





Dilated Causal Convolutions





Main Results of Health Forecasting Use cases with Proposed SCNN-BiLSTM

Use Case 1: Infectious Disease Forecasting

Infectious Disease Forecasting

Experimental Questions

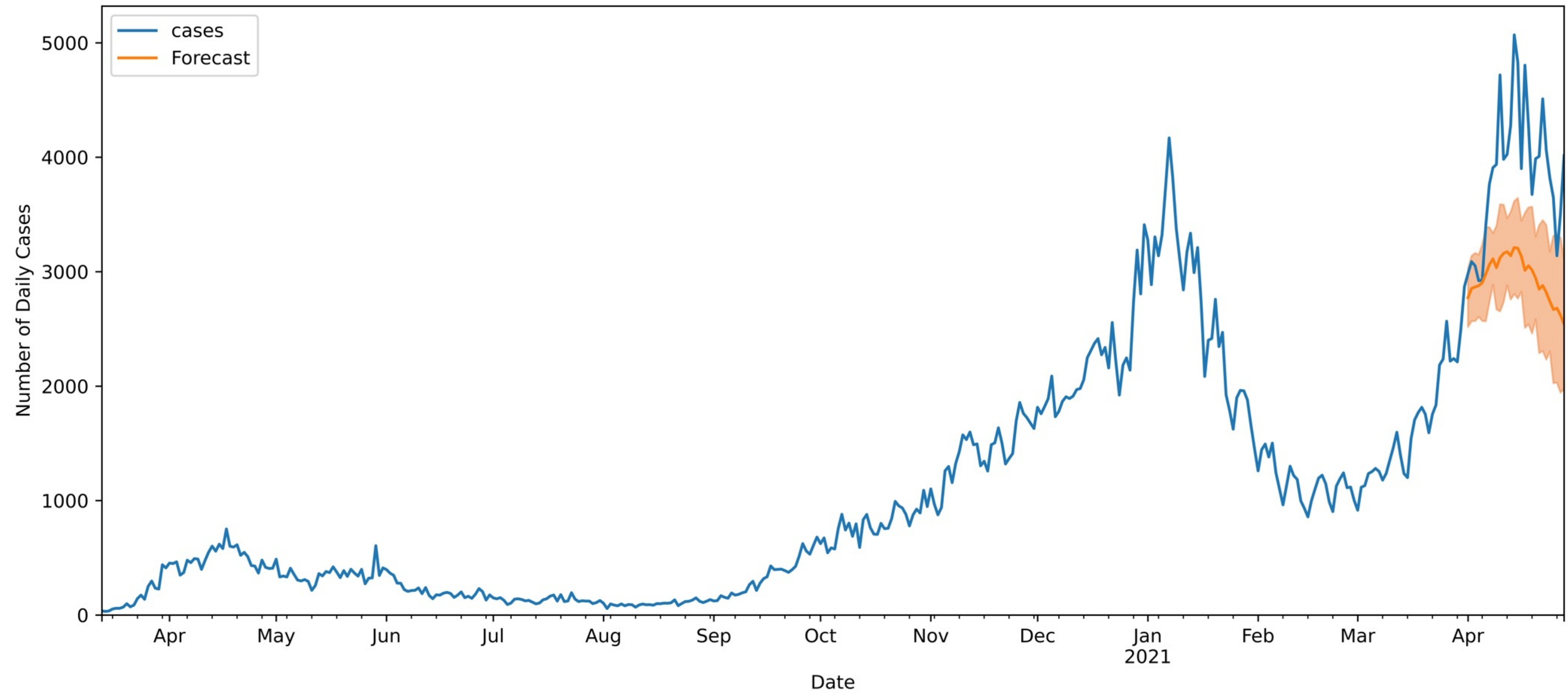
Forecast the spread of COVID-19 for the next 28 days in Ontario with multivariate data. Various configurations of the CNN and RNN were tested to determine whether:

1. The full history coverage provided by the SCNN performed better than a standard CNN.
2. The multivariate or univariate approach has superior performance.
3. The LSTM, BiLSTM or GRU was the optimal RNN for the hybrid model.

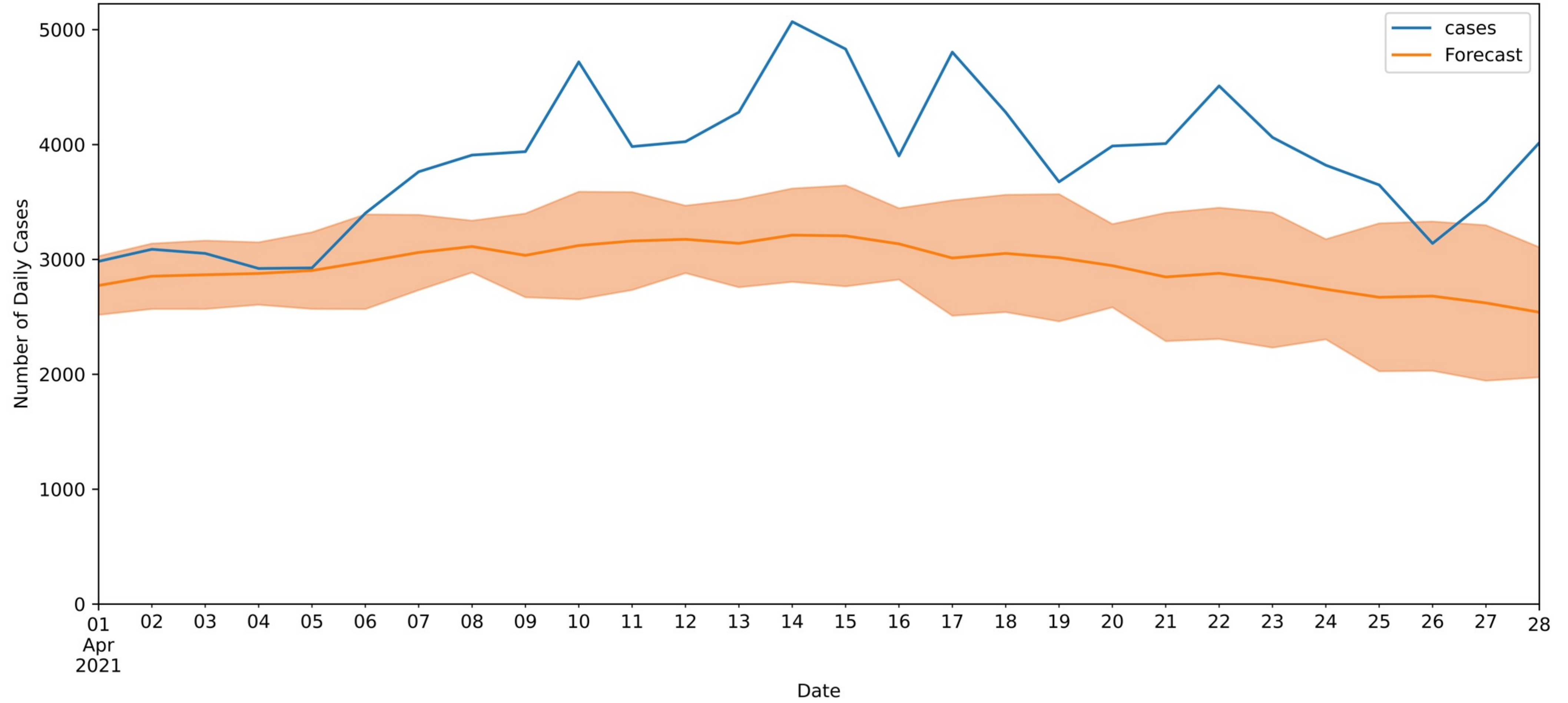
Forecasting results

Model	One-step (1 day)	Medium Horizon (1-14 days)	Long Horizon (15-28 days)
Univariate	7.043	16.878	27.917
Multivariate	4.928	14.924	27.528

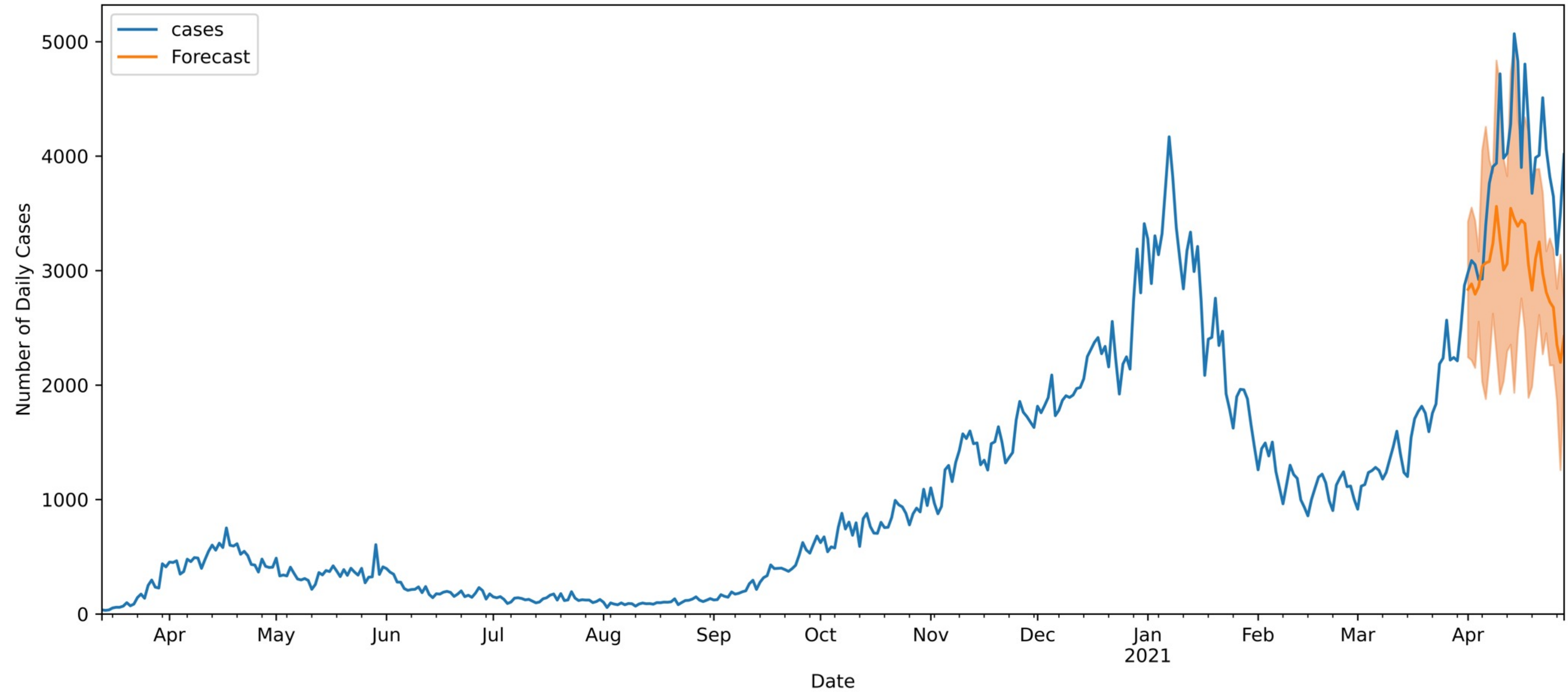
Forecasting with the Univariate SCNN-BiLSTM for Number of COVID-19 Infections (Complete Time Series)
MAPE Score: 22.39769



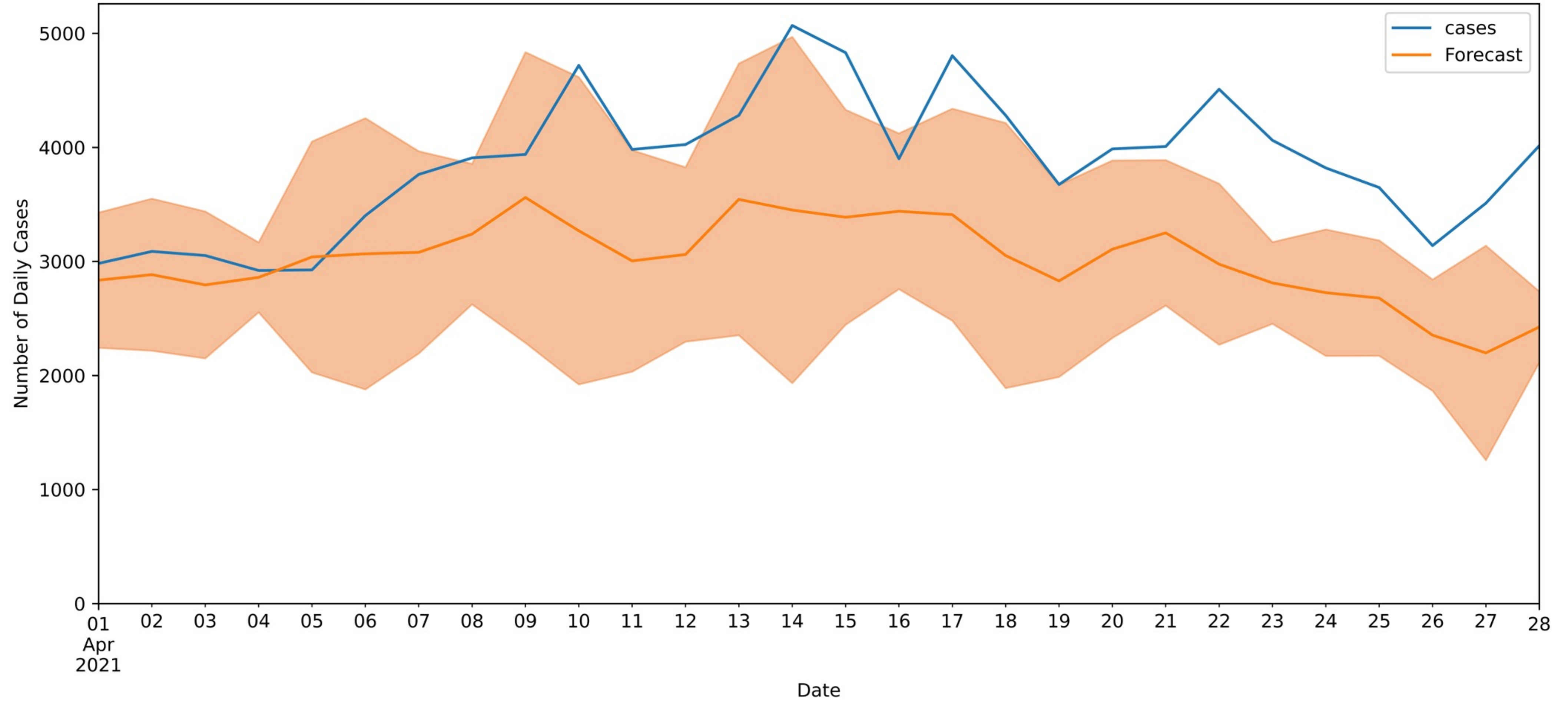
Forecasting with the Univariate SCNN-BiLSTM for Number of COVID-19 Infections (Forecast Time Peroid)



Forecasting with the Multivariate SCNN-BiLSTM for Number of COVID-19 Infections (Complete Time Series)
MAPE Score: 21.22596



Forecasting with the Multivariate SCNN-BiLSTM for Number of COVID-19 Infections (Forecast Time Peroid)



Infectious Disease Forecasting

Experimental Answers

The grid search revealed that:

1. That the SCNN outperformed standard CNN.
2. The multivariate was superior to the univariate approach, even in the presence of an incomplete dataset.
3. The BiLSTM was the optimal RNN for the hybrid model.

These results all support the effectiveness of the proposed multivariate SCNN-BiLSTM.

Use Case 2: Hospital Admissions Forecasting

Hospital Admissions Forecasting

Experimental Questions

Forecast the number of admissions for mental health patients using multivariate time-series data. Experiments were conducted to answer the following:

1. For one-step forecasting, does a statistical, neural network, or hybrid neural network have the best performance?
2. Does the full history coverage provided by the SCNN-BiLSTM provide an advantage compared to a standard CNN for multi-step forecasting?

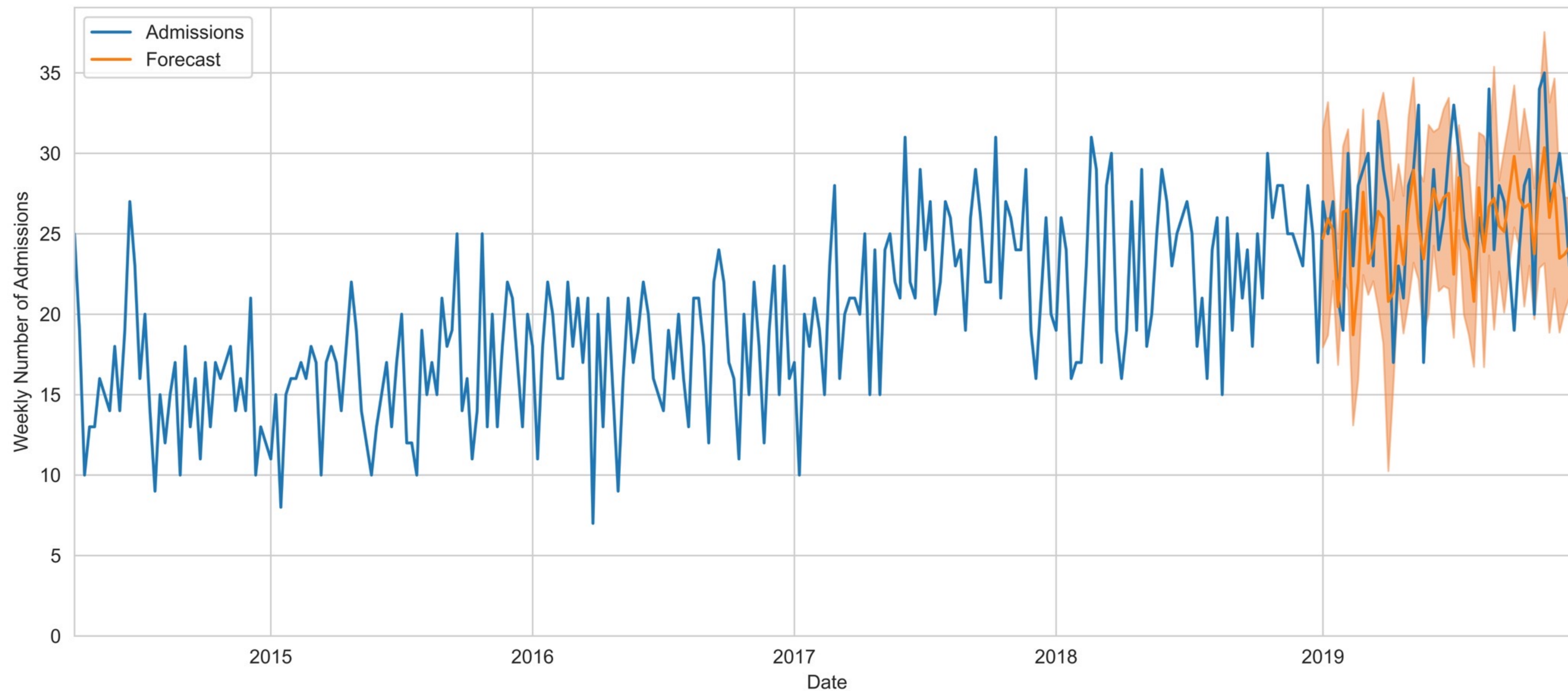
One-step and multi-step forecasting results

Model	Forecasting MAPE Score
$SARIMA(0, 1, 1)(3, 1, 0)_7$	15.5
$ARIMA(5, 1, 0)$	16.8
FCNN	15.3
CNN	14.9
LSTM	15.2
BiLSTM	15.5
CNN-LSTM	14.8
CNN-BiLSTM	13.8

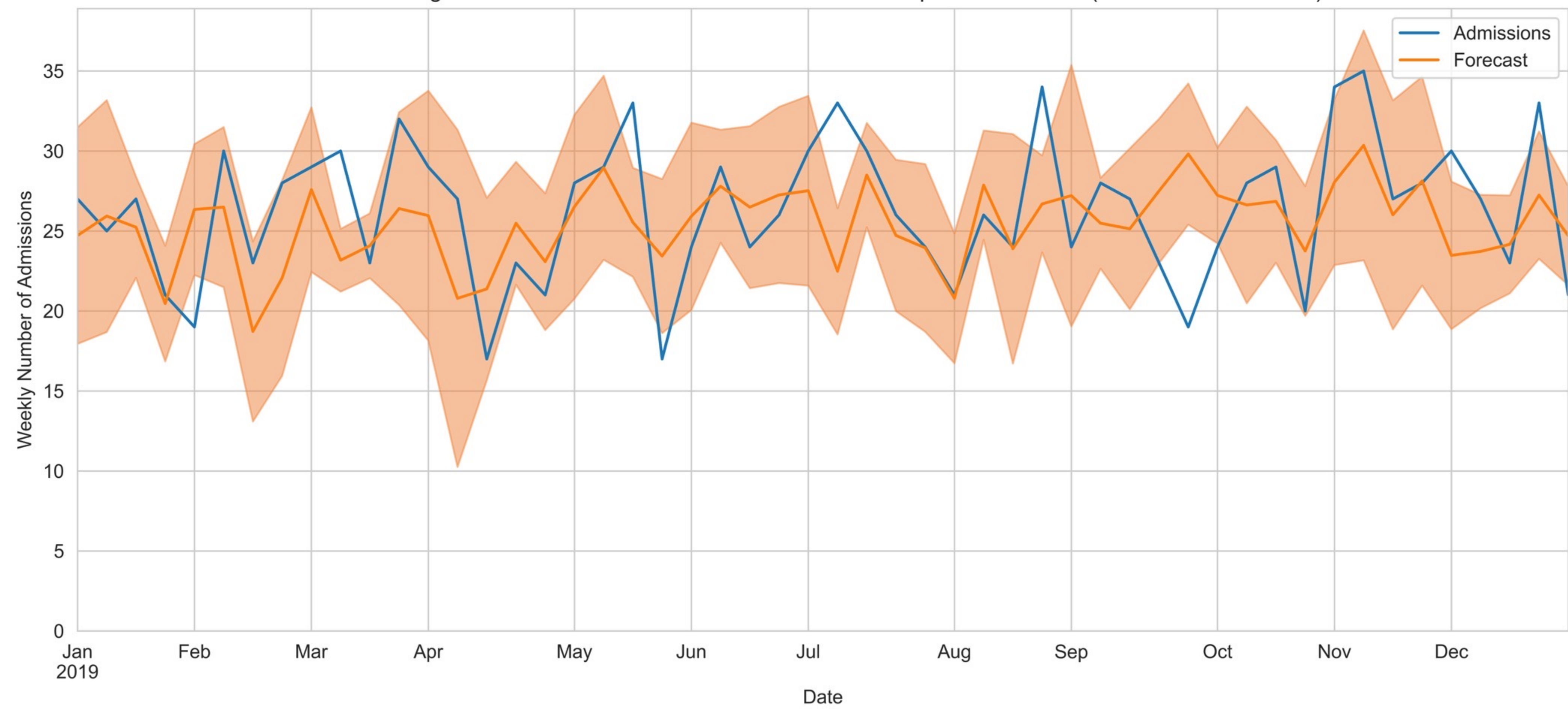
One-step and multi-step forecasting results

Model	Dataset	Forecasting MAPE Score
CNN-BiLSTM	Univariate	14.78
CNN-BiLSTM	Multivariate	13.33
SCNN-BiLSTM	Multivariate	12.96

Forecasting with the SCNN-BiLSTM for the Number of Hospital Admissions (Complete Time Series)
MAPE Score: 12.962



Forecasting with the SCNN-BiLSTM for the Number of Hospital Admissions (Forecast Time Peroid)



Hospital Admissions Forecasting

Experimental Answers

A grid search revealed the following:

1. The CNN-BiLSTM hybrid model outperformed various statistical and neural network techniques in the one-step forecast.
2. The SCNN-BiLSTM with full history coverage outperformed a standard CNN-BiLSTM hybrid model for multi-step forecasting.

Thank you for your time.

**I hope I have convinced you that Time Series
Forecasting is an exciting and important subject!**